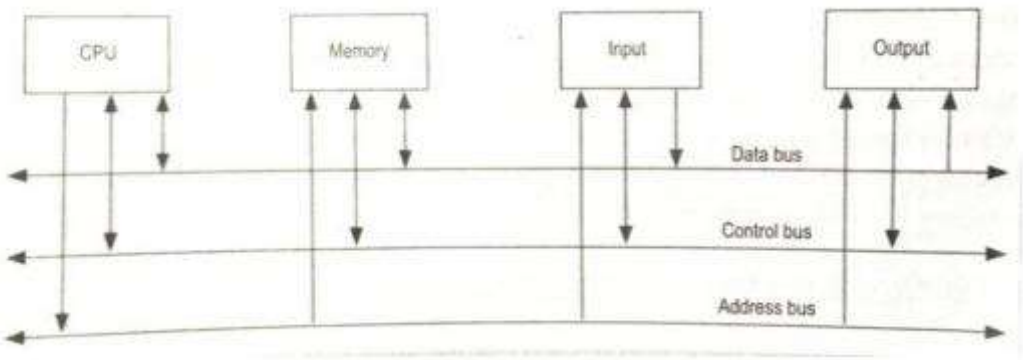
Chapter 1

Evolution of Microcomputer

Introduction

A microprocessor is a programmable electronics chip that has computing and decision making capabilities similar to central processing unit of a computer. Any microprocessor based systems having limited number of resources are called microcomputers. Nowadays, microprocessor can be seen in almost all types of electronics devices like mobile phones, printers, washing machines etc. Microprocessors are also used in advanced applications like radars, satellites and flights. Due to the rapid advancements in electronic industry and large scale integration of devices results in a significant cost reduction and increase application of microprocessors and their derivatives.



Microprocessor-based system

- Bit: A bit is a single binary digit.
- Word: A word refers to the basic data size or bit size that can be processed by the arithmetic and logic unit of the processor. A 16-bit binary number is called a word in a 16-bit processor.
- Bus: A bus is a group of wires/lines that carry similar information.
- System Bus: The system bus is a group of wires/lines used for communication between the microprocessor and peripherals.
- Memory Word: The number of bits that can be stored in a register or memory element is called a memory word.
- Address Bus: It carries the address, which is a unique binary pattern used to identify a memory location or an I/O port. For example, an eight bit address bus has eight lines and thus it can address 28 = 256 different locations. The locations in hexadecimal format can be written as 00H – FFH.
- Data Bus: The data bus is used to transfer data between memory and processor or between I/O device and processor. For example, an 8-bit processor will generally have an 8-bit data bus and a 16-bit processor will have 16-bit data bus.
- Control Bus: The control bus carry control signals, which consists of signals for selection of memory or I/O device from the given address, direction of data transfer and synchronization of data transfer in case of slow devices.

A typical microprocessor consists of arithmetic and logic unit (ALU) in association with control unit to process the instruction execution. Almost all the microprocessors are based on the principle of store-program concept. In store-program concept, programs or instructions are sequentially stored in the memory locations that are to be executed. To do any task using a microprocessor, it is to be programmed by the user. So the programmer must have idea about its internal resources, features and supported instructions. Each microprocessor has a set of instructions, a list which is provided by the microprocessor manufacturer. The instruction set of a microprocessor is provided in two forms: binary machine code and mnemonics.
Microprocessor communicates and operates in binary numbers 0 and 1. The set of instructions in the form of binary patterns is called a machine language and it is difficult for us to understand. Therefore, the binary patterns are given abbreviated names, called mnemonics, which forms the assembly language. The conversion of assembly-level language into binary machine-level language is done by using an application called assembler.

Technology Used:
The semiconductor manufacturing technologies used for chips are:

- Transistor-Transistor Logic (TTL)
- Emitter Coupled Logic (ECL)
- Complementary Metal-Oxide Semiconductor (CMOS)

Classification of Microprocessors:
Based on their specification, application and architecture microprocessors are classified.
Based on size of data bus:

- 4-bit microprocessor
- 8-bit microprocessor
- 16-bit microprocessor
- 32-bit microprocessor

Based on application:

- General-purpose microprocessor- used in general computer system and can be used by programmer for any application. Examples, 8085 to Intel Pentium.
- Microcontroller- microprocessor with built-in memory and ports and can be programmed for any generic control application. Example, 8051.
- Special-purpose processors- designed to handle special functions required for an application. Examples, digital signal processors and application-specific integrated circuit (ASIC) chips.

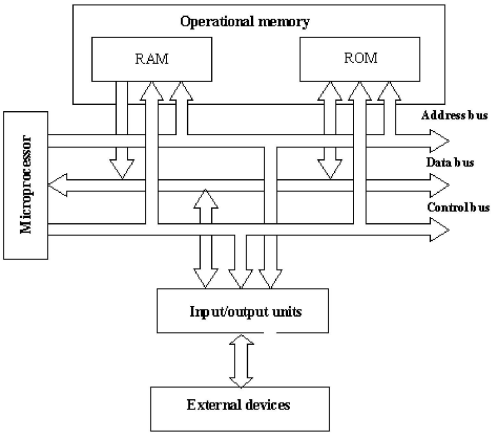
Based on architecture:

- Reduced Instruction Set Computer (RISC) processors
- Complex Instruction Set Computer (CISC) processors

The concept and architecture of a microcomputer

A microcomputer is a computer built on the basis of a microprocessor i.e. a processor implemented as an integrated circuit. Since all processors are now produced in the form of integrated circuits, we can

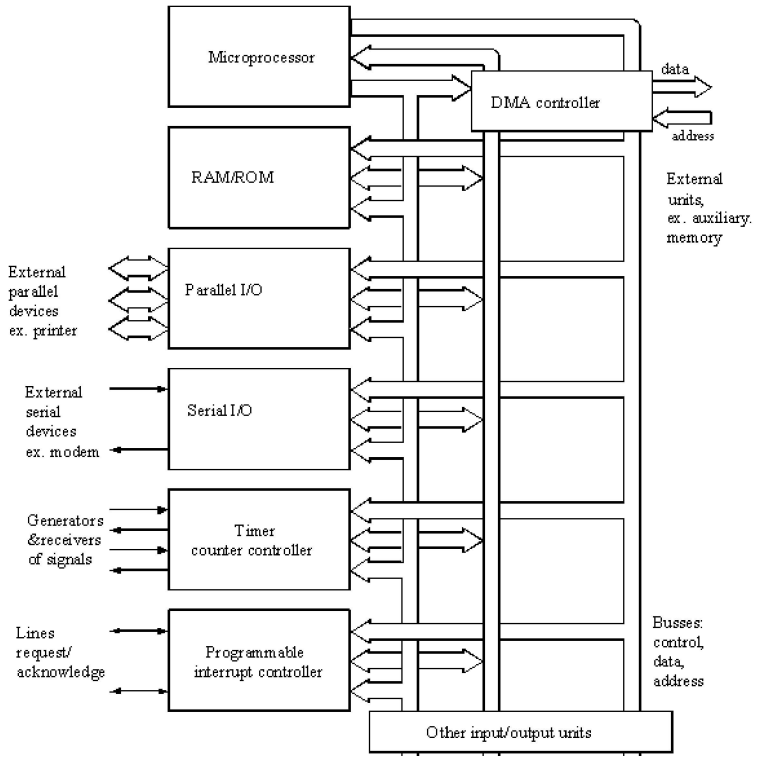
say that all computers are microcomputers. The general method for constructing microcomputers consists in connecting to the microprocessor busses additional sub-systems such as memories and peripheral device controllers (input/output units).
The basic block diagram of a simple microcomputer is shown in the figure below. We can see there a microprocessor with three its busses going out: data bus, address bus and control bus. To these busses, the following devices are connected: operational memory composed of RAM (Random Access Memory)and ROM (Read Only Memory) memories, as well as input/output units to which peripheral devices are connected.



Simple microcomputer

A more developed block diagram of a microcomputer is shown in the figure below. Besides RAM and ROM memories, more input/output units are connected to the microprocessor.
These input/output units include:

- parallel input/output controller in short - parallel I/O, parallel interface
- serial input/output controller in short - serial I/O, serial interface
- interrupt controller (handler)
- timer/counter controller
- Direct Memory Access (DMA) controller



General scheme of simple microcomputer

Parallel input/output controller maintains information exchange with peripheral devices, which send data in the parallel form. Examples of such devices are printers, display monitor, hard and floppy disk memories, keyboard. The activity of the controller is supervised by the microprocessor, which intervenes on each transfer of data by execution of respective instructions of data read or write from (to) the controller. The controller itself transfers data from its internal memory (registers) to peripheral devices.

The serial input/output controller maintains information exchange with peripheral devices, which send data in the serial form. Examples of such devices are a mouse and a modem for interconnections through telephone network. The controller implements in hardware conversion of serial data into their parallel form and vice versa (with the use of serial/parallel registers). The functioning of the controller is controlled by the microprocessor, which intervenes on each termination of data conversion by execution of respective data read or writes instructions.
Interrupt controller provides servicing interrupts coming in parallel from many external devices. Its task consists in receiving interrupt requests, registering them, performing selection to choose one which is to be serviced by the processor. The controller communicates with the processor to enable sending the identifier of the selected interrupt and to exchange control signals. The interrupt controller is supervised by the processor, which services the interrupt after receiving the interrupt from the controller.

The DMA controller enables parallel data exchange between external devices and the operational memory without involvement of the processor. This controller enables autonomous data transfers to/from operational memory. These transfers do not engage the processor, which can proceed with computations at the same time.

The Microprocessor's Impact on Society

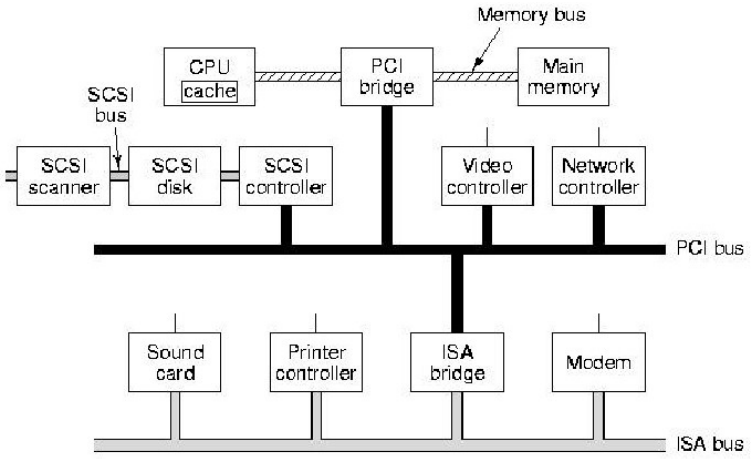
Today, we literally swim in a sea of silicon. There are dozens of chips in every car and dozens in every office. Microprocessors in the form of watches, pagers, cellular phones, and Sharp Wizard's adorn our bodies as jewelry. Beyond embedded systems, the rapid rise of the Internet now appears to be on the verge of ensuring that the personal computer truly becomes ubiquitous. Where in the past PCs have been concentrated in the workplace and scattered in the homes of the wealthier third of American society, the Internet is almost certain to lower the final obstacles and make it possible to create a new class of information appliances that will extend the reach of the Net until it matches that of television or the telephone. Microprocessors are everywhere, but how is modern life different? In many areas the advent of microelectronics obviously does have clear benefits. That can be seen in the effect of computing technology on the disabled; the transformation of certain kinds of boring and dangerous manual labor; advances in medical technology and weather forecasting. Indeed the easy answer is that microprocessors have transformed modern society. They affect the way we work and play, the way we travel and communicate. They offer remarkable processing power at infinitesimal cost.

Chapter 2

Architecture of Microprocessor

A bus is a collection of wires, chips and slots inside the computer through which data are transmitted from one part of the computer to another from peripheral devices. It is also called a pathway in the computer on which data travels. It is a set of parallel distinct wires, serving different purposes, which allow devices attached to it to communicate with the CPU.

The bus architecture in computer system is as shown below:



Bus architecture

The functions of BUS are:

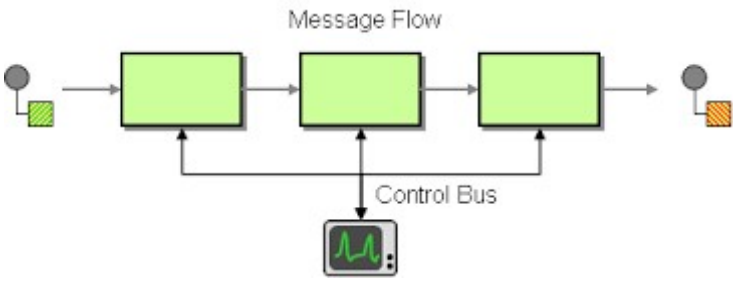
- It carries information from one component to another.
- It carries data, address or control signal.
- One component of the computer can interact with other through a bus.
- There are three main part of bus.

They are described below:

Control Bus:

It carries the control signal. The control signal is used for controlling and coordinating the various activities of the computer. It is generated from the control unit of the CPU. Different architectures result in a different number of lines within the control bus, as each line is used to perform a specific task.

For instance, different specific lines are used for each of read, write and reset requests. These are not a group of lines like address bus and data bus, but individual lines that provide a pulse to indicate a microprocessor operation. The control unit generates a specific control signal for every operation, such as memory read or input/output operation. This signal is also used to identify a device type, with which the microprocessor intends to communicate.



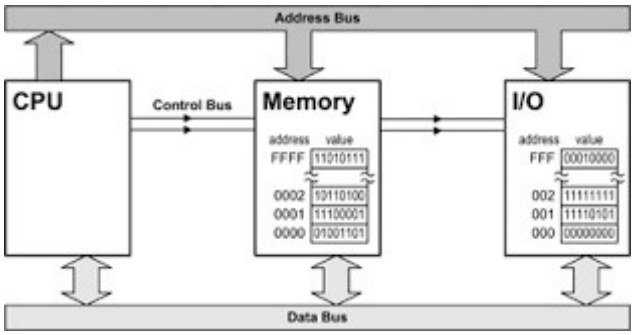
Control bus

Address Bus:

Address bus carries memory address within the device. It allows the CPU to reference memory locations within the device. It connects the CPU and other peripherals and carries only memory address. In a computer system, each peripheral or memory location is identified by a numerical value, called an address and the address bus is used to carry this numerical value as well as it also contains a few control lines to carry control commands. The address bus is unidirectional, bits flow in one direction from the processor to peripherals.

The processor uses the address bus to perform the first function of identifying a peripheral or a memory location. The address bus contains the connections between the processor and memory that carry the signals relating to the address which the CPU is processing at that time, such as the locations that the CPU is reading from or writing to. The processor uses the address bus to perform, identifying a peripheral or a memory location.

When the address bus carries 8 bit at a time, the CPU could address only 256 (i.e. 28) bytes of RAM. Most of the early PCs had 20 bit address buses. So, CPU could address 220bytes of data. Now, with 32-bit address buses, CPU can address 4GB of RAM. If there is wider bus path, more information can be processed at a time and hence, it also affects the processing speed of a computer.



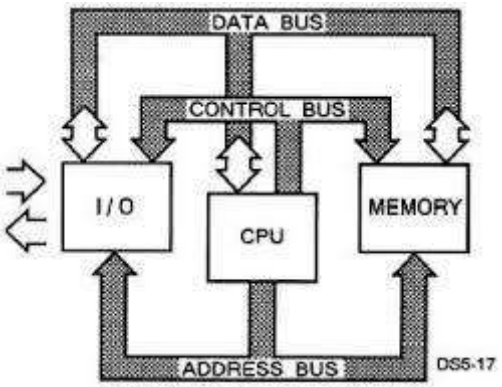
Address bus

Data Bus:

Data bus transfer data from one location to another across the computer. On these lines, the meaningful data which is to be retrieved from a device is placed. Data bus is used by CPU to transfer data. It may be 16-bit or 32-bit data bus. It is an electrical path that connects the CPU, memory and other hardware devices on the motherboard. These lines are bidirectional in which the data flows in both directions between the processor and memory and peripheral devices. The number of wires in the bus affects the speed at which data can be travel between hardware components just as the number of lanes on a highway affects the time it takes people to reach their destination. Each wire can transfer 1 bit of data at a time and 8 wire bus can move 8 bit at a time, which is 1-byte data at a time. A 16-bit bus can transfer 2

bytes. 32 bits can transfer 4 bytes and so on. Intel 80286 microprocessor used16 bit data bus and Intel 80386 used 32-bit data bus. When the data bus width grows larger, more data can be transferred.

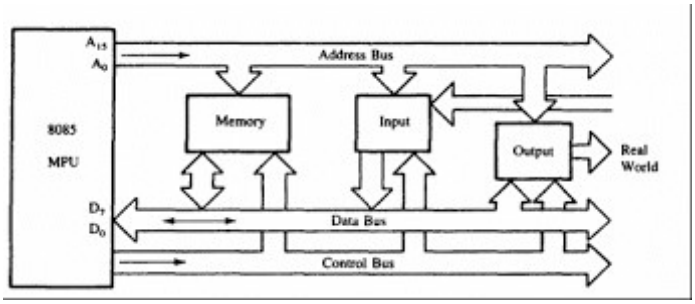
The transmission of the data on bus lines takes place between approximately 1M baud for the microcomputer to about 1000 M baud or more for the large more expensive computers (1 baud = 1 bit/sec). Communication between the different units of a processing system is carried out along address and data bus and also along various control lines. All control operations are governed by the master timing source and clock.



data bus

Bus organization of 8085

The microprocessor MPU performs various operations with peripheral devices or a memory location by using three sets of communication lines called buses: the address bus, the data bus and the control bus. And these three combined lines is called as system bus.



Bus structure

Address bus:

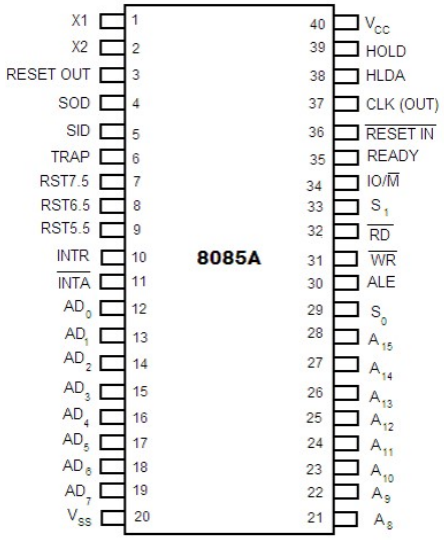
The address bus is a group of 16 lines generally called as A0 – A15 to carry a 16-bit address of memory location. In a computer system, each peripheral or memory location is identified by a binary number called an address. This is similar to the postal address of a house. The address bus is unidirectional, that means bit flow in only one direction from MPU to peripheral.

MPU carries 16-bit address i.e. $2^{16} = 65,536$ or 64K memory locations.

Data Bus:

The data bus is a group of eight bidirectional lines used for data flow in both the directions between MPH and peripheral devices.
The 8 data lines are manipulating 8-bit data ranging from 00 to FF i.e. ($2^8 = 256$) numbers from 0000 0000 -1111 1111
This 8 bit data is called as word length and the register size of a microprocessor and MPH is called 8–bit microprocessor.

Control bus:
Control bus is having various single lines used for sending control signals in the form of pulse to the memory and I/O devices.
The MPU generates specific control signals to perform a particular operations. Some of these control signals are memory read, memory write, I/O read and I/O write.



Pin diagram of 8085

The 8085 and Its Buses

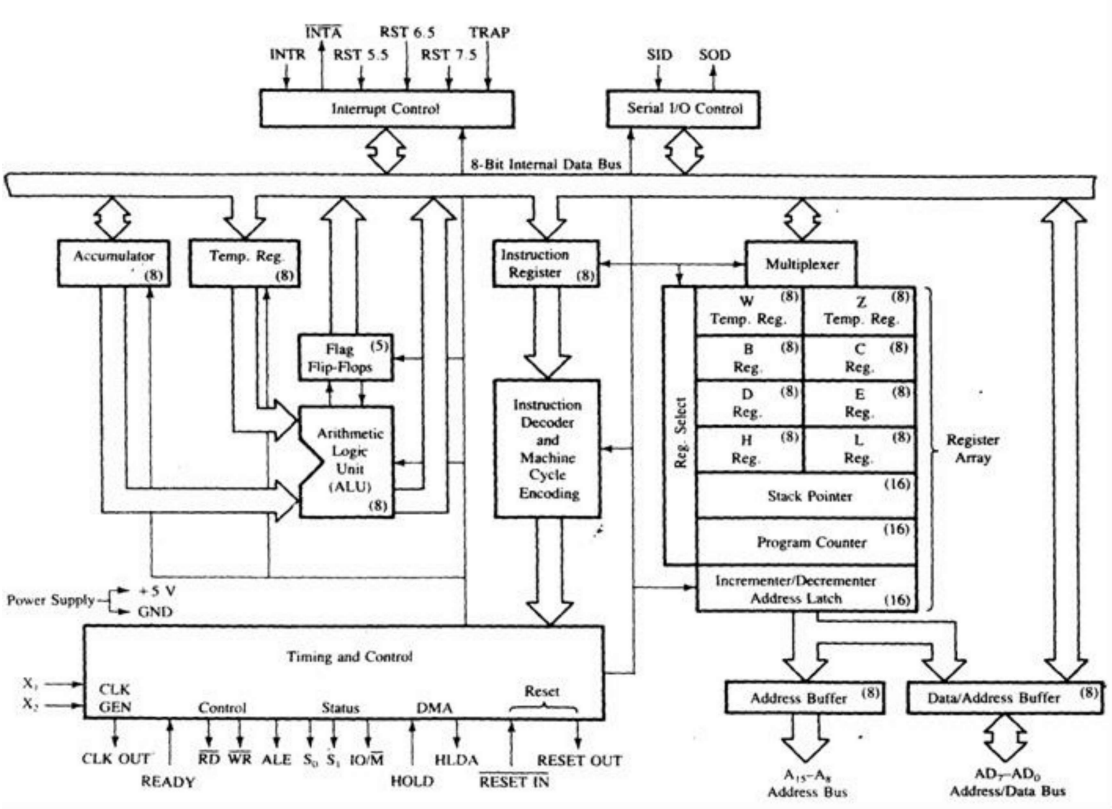
- The 8085 is an 8-bit general purpose microprocessor that can address 64K Byte of memory.
- It has 40 pins and uses +5V for power. It can run at a maximum frequency of 3 MHz.
- The pins on the chip can be grouped into 6 groups:
  - Address Bus.
  - Data Bus.
  - Control and Status Signals.
  - Power supply and frequency.
  - Externally Initiated Signals.
  - Serial I/O ports.

8085 Pin description

- Higher Order Address pins- $A_{15} – A_8$
  - The address bus has 8 signal lines A8 – A15 which are unidirectional.
- Lower Order Address/ Data Pins- AD_7-AD_0

    - These are time multiplexed pins and are DE-multiplexed using the pin ALE
    - So, the bits AD0 – AD7 are bi-directional and serve as A0 – A7 and D0 – D7 at the same time.
    - During the execution of the instruction, these lines carry the address bits during the early part, then during the late parts of the execution, they carry the 8 data bits.
    - In order to separate the address from the data, we can use a latch to save the value before the function of the bits changes.
- Control Pins – RD, WR
    - These are active low Read & Write pins
- Status Pins – ALE, IO/M (active low), S_1, S_0
    - ALE (Address Latch Enable)-Used to DE-multiplex AD_7-AD_0
    - IO/M – Used to select I/O or Memory operation
    - S_1,S_0 – Denote the status of data on data bus
- Interrupt Pins – TRAP, RST7.5, RST 6.5, RST 5.5, INTR, INTA
    - These are hardware interrupts used to initiate an interrupt service routine stored at predefined locations of the system memory.
- Serial I/O pins – SID (Serial Input Data), SOD (Serial Output Data)
    - These pins are used to interface 8085 with a serial device.
- Clock Pins- X_1, X_2, CLK(OUT)
    - X_1, X_2– These are clock input pins. A crystal is connected between these pins such that $f_{crystal} = 2f_{8085}$ where $f_{crystal}$= crystal frequency & f_{8085} = operating frequency of 8085
    - CLK(OUT) – This is an auxiliary clock output source
- Reset Pins – Reset In **(active low)**, Reset Out
    - **Reset In** is used to reset 8085 whereas **Reset Out** can be used to reset other devices in the system
- DMA (Direct Memory Access) pins – HOLD, HLDA
    - These pins are used when data transfer is to be performed directly between an external device and the main memory of the system.
- Power Supply Pins – $+V_{CC}$, V_{SS}

Architecture of 8085



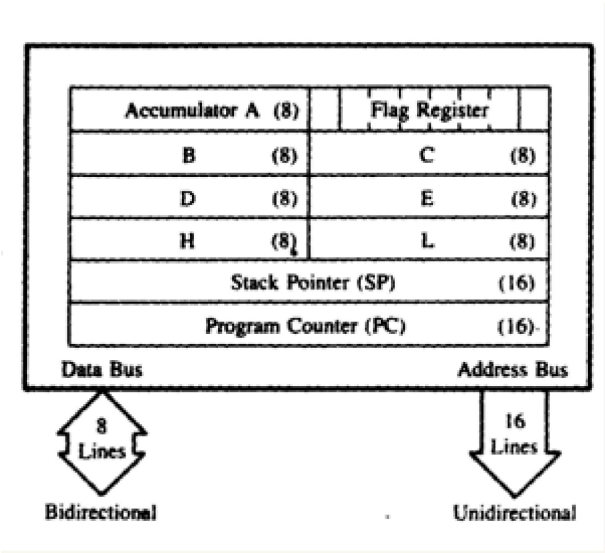
Function block diagram of 8085

This is the functional Block Diagram of 8085 Microprocessor.

Accumulator:-It is a 8-bit register which is used to perform arithmetical and logical operation. It stores the output of any operation. It also works as registers for i/o accesses.

Temporary Register:-It is a 8-bit register which is used to hold the data on which the accumulator is computing operation. It is also called as operand register because it provides operands to ALU.

Registers:-These are general purposes registers. Microprocessor consists 6 general purpose registers of 8-bit each named as B, C, D, E, H and L. Generally theses registers are not used for storing the data permanently. It carries the 8-bits data. These are used only during the execution of the instructions.
These registers can also be used to carry the 16 bits data by making the pair of 2 registers. The valid register pairs available are BC, DE HL. We cannot use other pairs except BC, DE and HL. These registers are programmed by user.



Programmatic model of 8085

ALU:-ALU performs the arithmetic operations and logical operation.
Flag Registers:-It consists of 5 flip flop which changes its status according to the result stored in an accumulator. It is also known as status registers. It is connected to the ALU.
There are five flip-flops in the flag register are as follows:

1. Sign(S)
2. zero(z)
3. Auxiliary carry(AC)
4. Parity(P)
5. Carry(C)

The bit position of the flip flop in flag register is:

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
S	Z		AC		P		CY

All of the three flip-flop set and reset according to the stored result in the accumulator.

1. **Sign-** If D7 of the result is 1 then sign flag is set otherwise reset. As we know that a number on the D7 always decides the sign of the number.
   If D7 is 1: the number is negative.
   If D7 is 0: the number is positive.
2. **Zeros (Z)-**If the result stored in an accumulator is zero then this flip flop is set otherwise it is reset.
3. **Auxiliary carry(AC)-**If any carry goes from D3 to D4 in the output then it is set otherwise it is reset
4. **Parity (P)-**If the no of 1's is even in the output stored in the accumulator then it is set otherwise it is reset for the odd
5. **Carry(C)-**If the result stored in an accumulator generates a carry in its final output then it is set otherwise it is reset.

Instruction registers (IR):- It is a 8-bit register. When an instruction is fetched from memory then it is stored in this register.

Instruction Decoder: - Instruction decoder identifies the instructions. It takes the information from instruction register and decodes the instruction to be performed.

Program Counter:- It is a 16 bit register used as memory pointer. It stores the memory address of the next instruction to be executed. So we can say that this register is used to sequencing the program. Generally the memory have 16 bit addresses so that it has 16 bit memory.
The program counter is set to 0000H.

Stack Pointer: - It is also a 16 bit register used as memory pointer. It points to the memory location called stack. Generally stack is a reserved portion of memory where information can be stores or taken back together.

Timing and Control Unit: - It provides timing and control signal to the microprocessor to perform the various operations. It has three control signals. It controls all external and internal circuits. It operates with reference to clock signals. It synchronizes all the data transfers.
There are three control signal:
1. ALE-Arithmetic Latch Enable, It provides control signal to synchronize the components of microprocessor.
2. RD- This is active low used for reading operation.
3. WR-This is active low used for writing operation.

There are three status signal used in microprocessor S0, S1 and IO/M. It changes its status according the provided input to these pins.

IO/M(Active Low)	S1	S2	Data Bus Status(Output)
0	0	0	Halt
0	0	1	Memory WRITE
0	1	0	Memory READ
1	0	1	IO WRITE
1	1	0	IO READ
0	1	1	Opcode fetch
1	1	1	Interrupt acknowledge

Serial Input Output Control-There are two pins in this unit. This unit is used for serial data communication.

Interrupt Unit-There are 6 interrupt pins in this unit. Generally an external hardware is connected to these pins. These pins provide interrupt signal sent by external hardware to microprocessor and microprocessor sends acknowledgement for receiving the interrupt signal. Generally INTA is used for acknowledgement.

Register Section:-Many registers has been used in microprocessor.
It consists of PIPO (Parallel Input Parallel Output) register.

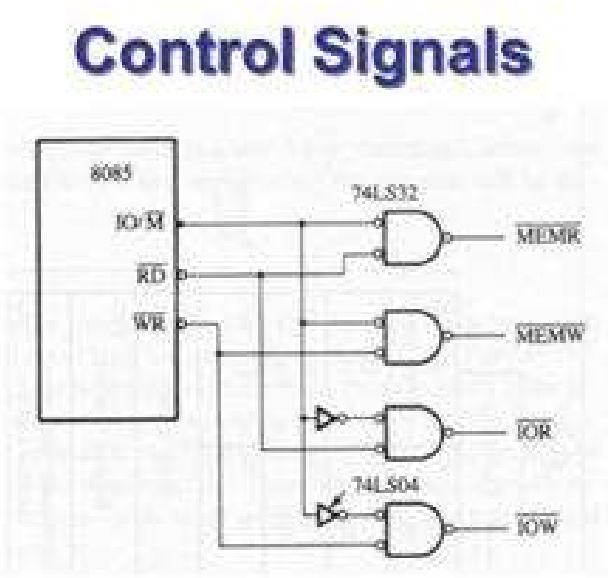
What is an Opcode?

Opcode is nothing but the machine language instruction which denotes the microprocessor about what operation should be performed on the specific data. Microprocessor converts the instruction into suitable machine language, so that it can understand the operation to be performed and executes it.

Consider MVI A, 18H.

When the above instruction is to be executed, the microprocessor gets the Opcode for MVI A and performs the necessary operation on the data which is 18H in this case. The Opcode for MVI A is 3EH. So the microprocessor first reads this Opcode from the instruction and then performs the operation specified by Opcode over the data given.

Now let us assume we want to store the above instruction in a specific address say 5500H. We know that in 8085 processor only one byte can be stored in each address location. Therefore the Opcode 3EH is stored at the location 5500H and the data 18H is stored at the next location 5501H.



Schematic diagram of control signal generation

READ Operation:

- During T1 state, microprocessor uses IO/M(bar), S0, S1 signals are used to instruct microprocessor to fetch opcode.
- Thus when IO/M(bar)=0, S0=S1= 1, it indicates opcode fetch operation.
- During this operation 8085 transmits 16-bit address and also uses ALE signal for address latching.
- At T2 state microprocessor uses read signal and make data ready from that memory location to read opcode from memory and at the same time program counter increments by 1 and points next instruction to be fetched.
- In this state microprocessor also checks READY input signal, if this pin is at low logic level ie. '0' then microprocessor adds wait state immediately between T2 and T3.

- At T3, microprocessor reads opcode and stores it into instruction register to decode it further.
- During T4 microprocessor performs internal operation like decoding opcode and providing necessary actions.
- The opcode is decoded to know whether T5 or T6 states are required, if they are not required then 碌p performs next operation.

WRITE Operation:

- It is used to fetch one byte from the memory.
- It requires 3 T-States.
- It can be used to fetch operand or data from the memory.
- During T1, A8-A15 contains higher byte of address. At the same time ALE is high. Therefore Lower byte of address A0-A7 is selected from AD0-AD7.
- Since it is memory ready operation, IO/M(bar) goes low.
- During T2 ALE goes low, RD(bar) goes low. Address is removed from AD0-AD7 and data D0-D7 appears on AD0-AD7.
- During T3, Data remains on AD0-AD7 till RD(bar) is at low signal.

Steps to execute a stored program

Two steps are performed that are fetch operation and execution operation.

Fetch operation: -

1. microprocessor will transfer 16 bit address from PC to address line to select desired location
2. Microprocessor gives reading location so 8 bit instruction is transferred from selected memory location to data pin
3. instruction code is transferred by mi9croprocessor to instruction register from data bus

Execution operation:-

1. Instruction code is transferred from IR to instruction decoder

2. The instruction code as of 2 or 3 bytes when 1st ckt. Will read 2nd and 3rd byte of instruction from memory and then execute corresponding instruction.

Chapter 3

Instruction Timing and Cycle

An instruction cycle (also known as the fetch–decode–execute cycle or the fetch-execute cycle) is the basic operational process of a computer. It is the process by which a computer retrieves a program instruction from its memory, determines what actions the instruction dictates, and carries out those actions. This cycle is repeated continuously by a computer's central processing unit (CPU), from boot-up to when the computer is shut down.

In simpler CPU's the instruction cycle is executed sequentially, each instruction being processed before the next one is started. In most modern CPU's the instruction cycles are instead executed concurrently, and often in parallel, through an instruction pipeline: the next instruction starts being processed before the previous instruction has finished, which is possible because the cycle is broken up into separate steps.

1. **Initiating the cycle**
   - The cycle starts immediately when power is applied to the system using an initial PC value that is predefined for the system architecture
   - Typically this address points to instructions in a read-only memory (ROM) (not the random access memory or RAM) which begins the process of loading the operating system.

2. **Fetch the Instruction**

   Step 1 of the Instruction Cycle is called the Fetch Cycle. This step is the same for each instruction.

   i. The CPU sends PC to the MAR and sends a READ command on the control bus
   ii. In response to the read command (with address equal to PC), the memory returns the data stored at the memory location indicated by PC on the data bus.
   iii. The CPU copies the data from the data bus into its MDR (also known as MBR)
   iv. A fraction of a second later, the CPU copies the data from the MDR to the Instruction Register (IR)
   v. The PC is incremented so that it points to the following instruction in memory. This step prepares the CPU for the next cycle. The Control Unit fetches the instruction's address from the Memory Unit

3. **Decode the Instruction**
   - Step 2 of the instruction Cycle is called the Decode Cycle. The decoding process allows the CPU to determine what instruction is to be performed, so that the CPU can tell how many operands it needs to fetch in order to perform the instruction.
   - The opcode fetched from the memory is decoded for the next steps and moved to the appropriate registers. The decoding is done by the CPU's Control Unit.

4. **Read the effective address**
   - Step 3 is deciding which operation it is. If this is a Memory operation - in this step the computer checks if it's a direct or indirect memory operation:
   - Direct memory instruction - Nothing is being done.
   - Indirect memory instruction - The effective address is being read from the memory. If this is a I/O or Register instruction - the computer checks its kind and executes the instruction.

5. **Execute the Instruction**
   - Step 4 of the Instruction Cycle is the Execute Cycle. Here, the function of the instruction is performed.
   - If the instruction involves arithmetic or logic, the Arithmetic Logic Unit is utilized. This is the only stage of the instruction cycle that is useful from the perspective of the end user.
   - Everything else is overhead required to make the execute stage happen.

Machine cycle

A *machine cycle*, also called a *processor cycle* or a *instruction cycle*, is the basic operation performed by a central processing unit (CPU). A CPU is the main logic unit of a computer.

A machine cycle consists of a sequence of three steps that is performed continuously and at a rate of millions per second while a computer is in operation. They are *fetch*, *decode* and *execute*. There also is a fourth step, *store*, in which input and output from the other three phases is stored in memory for later use; however, no actual processing is performed during this step.

In the fetch step, the *control unit* requests that main memory provide it with the instruction that is stored at the *address* (i.e., location in memory) indicated by the control unit's *program counter*.

The control unit is a part of the CPU that also decodes the instruction in the *instruction register*. A *register* is a very small amount of very fast memory that is built into the CPU in order to speed up its operations by providing quick access to commonly used values; instruction registers are registers that hold the instruction being executed by the CPU. Decoding the instructions in the instruction register involves breaking the *operand field* into its components based on the instructions *opcode*.

Opcode (an abbreviation of *operation code*) is the portion of a *machine language* instruction that specifies what operation is to be performed by the CPU. Machine language, also called *machine code*, refers to instructions coded in patterns of bits (i.e., zeros and ones) that are directly readable and executable by a CPU.

A program counter, also called the *instruction pointer* in some computers, is a register that indicates where the computer is in its instruction sequence. It holds either the address of the instruction currently being executed or the address of the next instruction to be executed, depending on the details of the particular computer. The program counter is automatically incremented for each machine cycle so that instructions are normally retrieved sequentially from memory.

The control unit places these instructions into its instruction register and then increments the program counter so that it contains the address of the next instruction stored in memory. It then executes the instruction by activating the appropriate circuitry to perform the requested task. As soon as the instruction has been executed, it restarts the machine cycle, beginning with the fetch step.

T states

One complete cycle of clock is called as T-state as shown in the above figure. A T-state is measured from the falling edge of one clock pulse to the falling edge of the next clock pulse. Various versions of 8086 have maximum clock frequency from 5MHz to 10MHz. Hence the minimum time for one T-state is between 100 to 200 nsec.

Instruction Execution And Timing Diagram:

Each instruction in 8085 microprocessor consists of two part- operation code (opcode) and operand. The opcode is a command such as ADD and the operand is an object to be operated on, such as a byte or the content of a register.

Instruction Cycle: The time taken by the processor to complete the execution of an instruction. An instruction cycle consists of one to six machine cycles.

Machine Cycle: The time required to complete one operation; accessing either the memory or I/O device. A machine cycle consists of three to six T-states.

T-State: Time corresponding to one clock period. It is the basic unit to calculate execution of instructions or programs in a processor.

To execute a program, 8085 performs various operations as:

- Opcode fetch
- Operand fetch
- Memory read/write
- I/O read/write

External communication functions are:

- Memory read/write
- I/O read/write
- Interrupt request acknowledge
-

Opcode Fetch Machine Cycle:

It is the first step in the execution of any instruction. The timing diagram of this cycle is given below

The following points explain the various operations that take place and the signals that are changed during the execution of opcode fetch machine cycle:

T1 clock cycle

i. The content of PC is placed in the address bus; AD0 - AD7 lines contains lower bit address and A8 – A15 contains higher bit address.
ii. M IO/ signal is low indicating that a memory location is being accessed. S1 and S0 also changed to the levels as indicated in Table 1.
iii. ALE is high, indicates that multiplexed AD0 – AD7 act as lower order bus.

T2 clock cycle

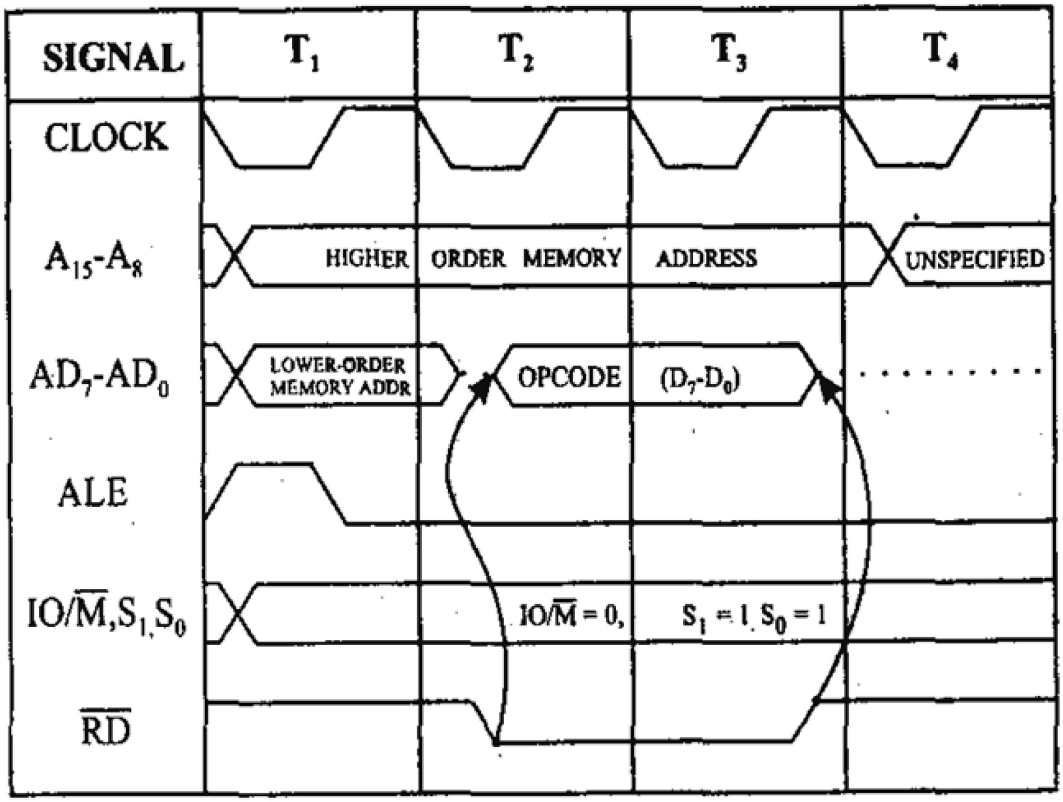
i. Multiplexed address bus is now changed to data bus.
ii. The RD signal is made low by the processor. This signal makes the memory device load the data bus with the contents of the location addressed by the processor.

T3 clock cycle

i. The opcode available on the data bus is read by the processor and moved to the instruction register. ii. The RD signal is deactivated by making it logic 1.

T4 clock cycle

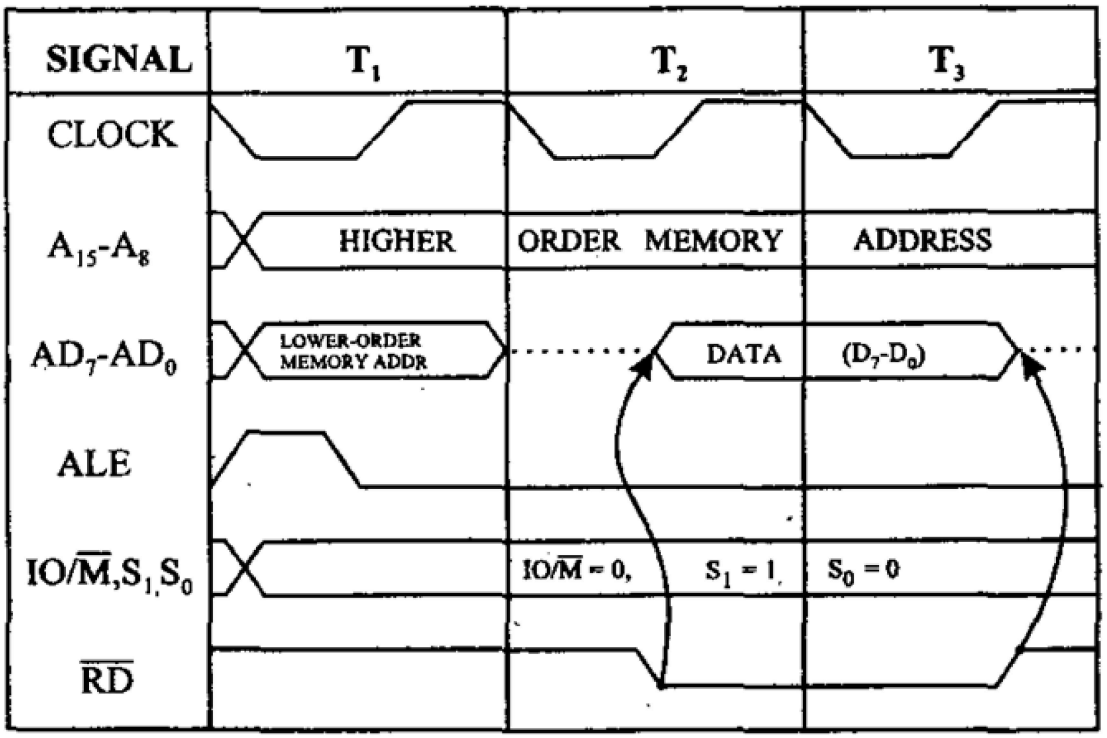
i. The processor decode the instruction in the instruction register and generate the necessary control signals to execute the instruction. Based on the instruction further operations such as fetching, writing into memory etc takes place.



Timing diagram for opcode fetch cycle

Memory Read Machine Cycle:

The memory read cycle is executed by the processor to read a data byte from memory. The machine cycle is exactly same to opcode fetch except: a) It has three T-states b) The S0 signal is set to 0. The timing diagram of this cycle is given in Fig.



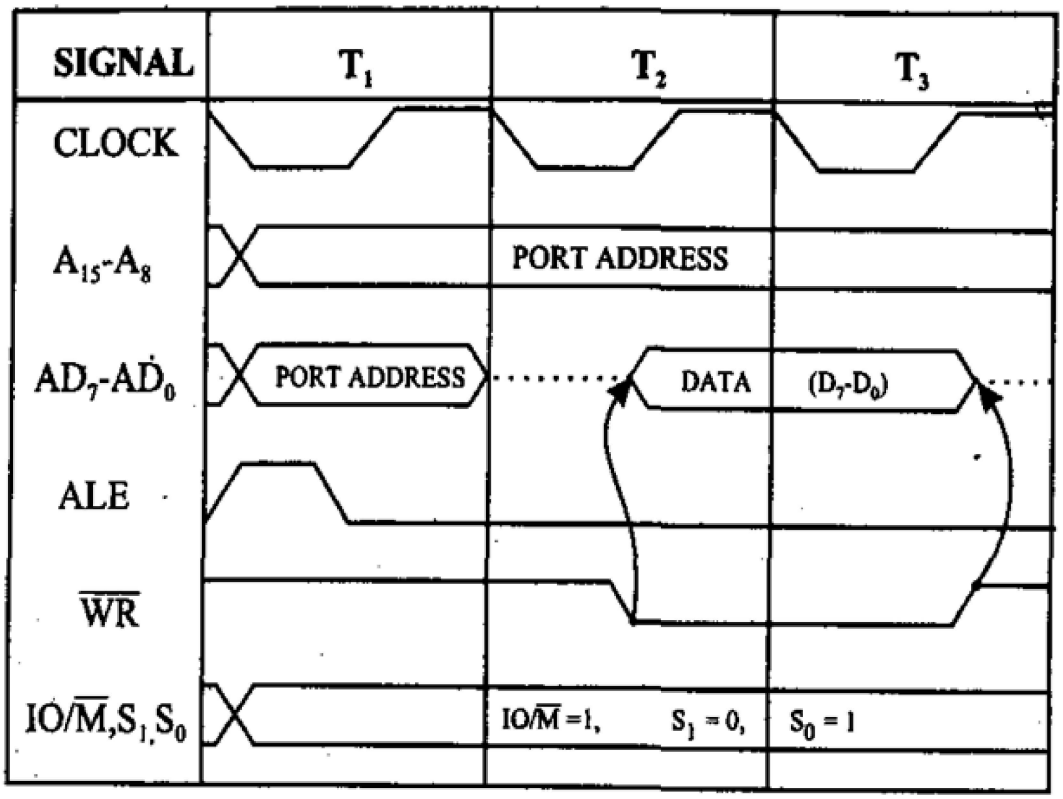
Timing diagram for memory read machine cycle

Memory Write Machine Cycle:

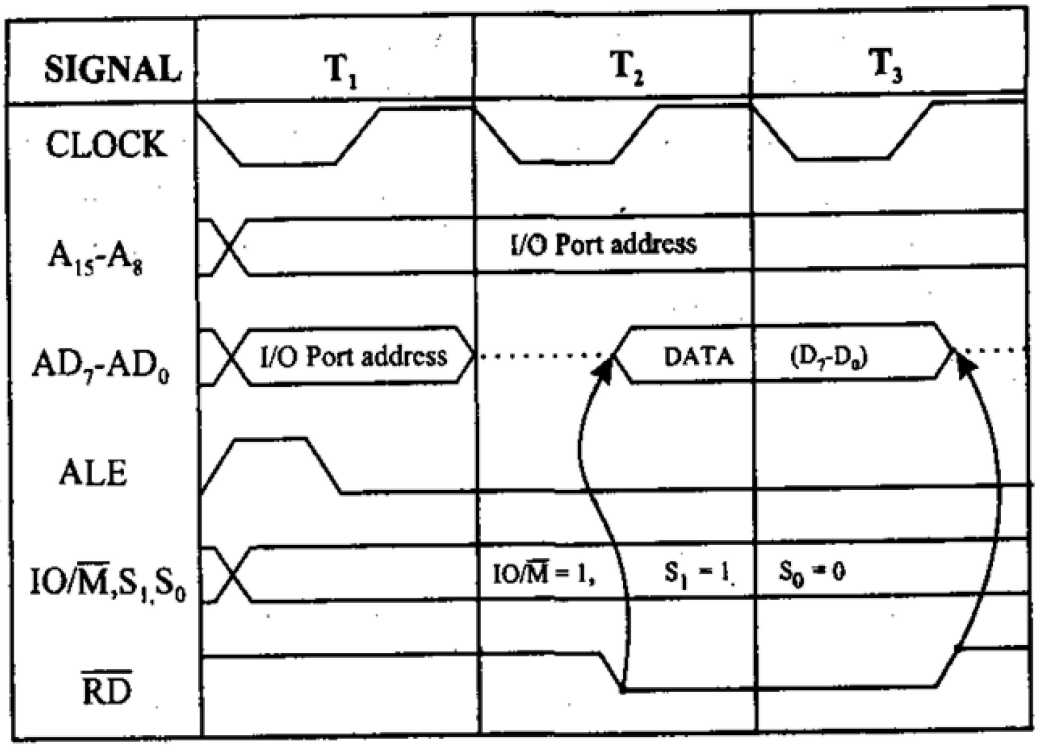
The memory write cycle is executed by the processor to write a data byte in a memory location. The processor takes three T-states and WR signal is made low. The timing diagram of this cycle is given below.

I/O Read Cycle:

The I/O read cycle is executed by the processor to read a data byte from I/O port or from peripheral, which is I/O mapped in the system. The 8-bit port address is placed both in the lower and higher order address bus. The processor takes three T-states to execute this machine cycle. The timing diagram of this cycle is given below



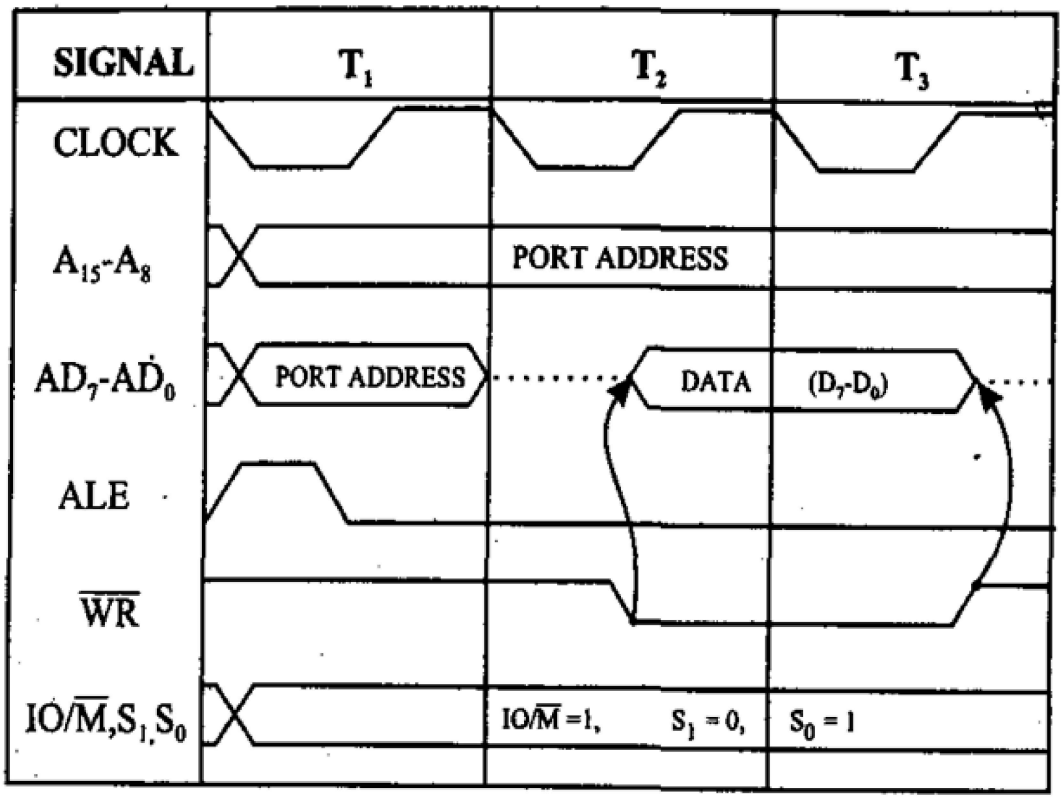
Timing diagram for memory write machine cycle



Timing diagram I/O read machine cycle

I/O Write Cycle:

The I/O write cycle is executed by the processor to write a data byte to I/O port or to a peripheral, which is I/O mapped in the system. The processor takes three T-states to execute this machine cycle. The timing diagram of this cycle is given in dia.



Timing diagram I/O write machine cycle

Ex: Timing diagram for IN 80H.

Chapter – 4

Programming (with respect to8085 microprocessor)

Assembly languages

An assembly (or assembler) language, often abbreviated asm, is a low-level programming language for a computer, or other programmable device, in which there is a very strong (but often not one-to-one) correspondence between the language and the architecture's machine code instructions. Each assembly language is specific to a particular computer architecture. In contrast, most high-level programming languages are generally portable across multiple architectures but require interpreting or compiling. Assembly language may also be called symbolic machine code.

Machine language

Machine code or machine language is a set of instructions executed directly by a computer's central processing unit (CPU). Each instruction performs a very specific task, such as a load, a jump, or an ALU operation on a unit of data in a CPU register or memory. Every program directly executed by a CPU is made up of a series of such instructions. (The phrase 'directly executed' needs some clarification; machine code is by definition the lowest level of programming detail visible to the programmer, but internally many processors use microcode or optimize and transform machine code instructions into sequences of micro-ops in a sophisticated way.)

Mnemonics

Mnemonics allow users to access quickly a wide variety of commands, services, programs and functions without the need to type out extended phrases. One example of a mnemonic code is the term "inc," which on an Intel microprocessor refers to the command "increase by one." Rather than type the entire phrase, the letters "inc" can be entered. Mnemonic code derives from the concept of traditional mnemonics in which abbreviations, rhymes or simple stories are used to help people remember information.

Instruction Set and Execution In 8085

Based on the design of the ALU and decoding unit, the microprocessor manufacturer provides instruction set for every microprocessor. The instruction set consists of both machine code and mnemonics.

An instruction is a binary pattern designed inside a microprocessor to perform a specific function. The entire group of instructions that a microprocessor supports is called instruction set. Microprocessor instructions can be classified based on the parameters such functionality, length and operand addressing.

Classification based on functionality:

I. **Data transfer operations:** This group of instructions copies data from source to destination. The content of the source is not altered.
II. **Arithmetic operations:** Instructions of this group perform operations like addition, subtraction, increment & decrement. One of the data used in arithmetic operation is stored in accumulator and the result is also stored in accumulator.
III. **Logical operations:** Logical operations include AND, OR, EXOR, NOT. The operations like AND, OR and EXOR uses two operands, one is stored in accumulator and other can be any register or

memory location. The result is stored in accumulator. NOT operation requires single operand, which is stored in accumulator.

IV. **Branching operations:** Instructions in this group can be used to transfer program sequence from one memory location to another either conditionally or unconditionally. V. Machine control operations: Instruction in this group control execution of other instructions and control operations like interrupt, halt etc.

Classification based on length:

I. **One-byte instructions:** Instruction having one byte in machine code. Examples are depicted in Table 2

II. **Two-byte instructions:** Instruction having two byte in machine code. Examples are depicted in Table 3.

III. **Three-byte instructions:** Instruction having three byte in machine code. Examples are depicted in Table 4.

Table 2 Examples of one byte instructions

Opcode	Operand	Machine code/Hex code
MOV	A, B	78
ADD	M	86

Table 3 Examples of two byte instructions

Opcode	Operand	Machine code/Hex code	Byte description
MVI A,	7FH	3E	First byte
		7F	Second byte
ADI	0FH	C6	First byte
		0F	Second byte

Table 4 Examples of three byte instructions

Opcode	Operand	Machine code/Hex code	Byte description
JMP	9050H	C3	First byte
		50	Second byte
		90	Third byte
LDA	8850H	3A	First byte
		50	Second byte
		88	Third byte

Addressing Modes in Instructions:

The process of specifying the data to be operated on by the instruction is called addressing. The various formats for specifying operands are called addressing modes.

The 8085 has the following five types of addressing:

I. Immediate addressing

II. Memory direct addressing
III. Register direct addressing
IV. Indirect addressing
V. Implicit addressing

Immediate Addressing:
In this mode, the operand given in the instruction - a byte or word – transfers to the destination register or memory location.
Ex: MVI A, 9AH

- The operand is a part of the instruction.
- The operand is stored in the register mentioned in the instruction.

Memory Direct Addressing:
Memory direct addressing moves a byte or word between a memory location and register. The memory location address is given in the instruction.
Ex: LDA 850FH
This instruction is used to load the content of memory address 850FH in the accumulator.

Register Direct Addressing:
Register direct addressing transfer a copy of a byte or word from source register to destination register.
Ex: MOV B, C
It copies the content of register C to register B.

Indirect Addressing:
Indirect addressing transfers a byte or word between a register and a memory location.
Ex: MOV A, M
Here the data is in the memory location pointed to by the contents of HL pair. The data is moved to the accumulator.

Implicit Addressing
In this addressing mode the data itself specifies the data to be operated upon.
Ex: CMA
The instruction complements the content of the accumulator. No specific data or operand is mentioned in the instruction.

INSTRUCTION SET OF 8085

Data Transfer Instructions

Opcode	Operand	Description
Copy from source to destination		
MOV	Rd, Rs M, Rs Rd, M	This instruction copies the contents of the source register into the destination register; the contents of the source register are not altered. If one of the operands is a memory location, its location is specified by the contents of the HL registers. Example: MOV B, C or MOV B, M
Move immediate 8-bit		
MVI	Rd, data M, data	The 8-bit data is stored in the destination register or memory. If the operand is a memory location, its location is specified by the contents of the HL registers. Example: MVI B, 57H or MVI M, 57H
Load accumulator		
LDA	16-bit address	The contents of a memory location, specified by a 16-bit address in the operand, are copied to the accumulator. The contents of the source are not altered. Example: LDA 2034H
Load accumulator indirect		
LDAX	B/D Reg. pair	The contents of the designated register pair point to a memory location. This instruction copies the contents of that memory location into the accumulator. The contents of either the register pair or the memory location are not altered. Example: LDAX B
Load register pair immediate		
LXI	Reg. pair, 16-bit data	The instruction loads 16-bit data in the register pair designated in the operand. Example: LXI H, 2034H or LXI H, XYZ
Load H and L registers direct		
LHLD	16-bit address	The instruction copies the contents of the memory location pointed out by the 16-bit address into register L and copies the contents of the next memory location into register H. The contents of source memory locations are not altered. Example: LHLD 2040H

Store accumulator direct

STA	16-bit address	The contents of the accumulator are copied into the memory location specified by the operand. This is a 3-byte instruction, the second byte specifies the low-order address and the third byte specifies the high-order address. Example: STA 4350H

Store accumulator indirect

STAX	Reg. pair	The contents of the accumulator are copied into the memory location specified by the contents of the operand (register pair). The contents of the accumulator are not altered. Example: STAX B

Store H and L registers direct

SHLD	16-bit address	The contents of register L are stored into the memory location specified by the 16-bit address in the operand and the contents of H register are stored into the next memory location by incrementing the operand. The contents of registers HL are not altered. This is a 3-byte instruction, the second byte specifies the low-order address and the third byte specifies the high-order address. Example: SHLD 2470H

Exchange H and L with D and E

XCHG	None	The contents of register H are exchanged with the contents of register D, and the contents of register L are exchanged with the contents of register E. Example: XCHG

Copy H and L registers to the stack pointer

SPHL	None	The instruction loads the contents of the H and L registers into the stack pointer register, the contents of the H register provide the high-order address and the contents of the L register provide the low-order address. The contents of the H and L registers are not altered. Example: SPHL

Exchange H and L with top of stack

XTHL	None	The contents of the L register are exchanged with the stack location pointed out by the contents of the stack pointer register. The contents of the H register are exchanged with the next stack location (SP+1); however, the contents of the stack pointer register are not altered. Example: XTHL

Push register pair onto stack

PUSH	Reg. pair	The contents of the register pair designated in the operand are copied onto the stack in the following sequence. The stack pointer register is decremented and the contents of the high-order register (B, D, H, A) are copied into that location. The stack pointer register is decremented again and the contents of the low-order register (C, E, L, flags) are copied to that location. Example: PUSH B or PUSH A

Pop off stack to register pair

POP	Reg. pair	The contents of the memory location pointed out by the stack pointer register are copied to the low-order register (C, E, L, status flags) of the operand. The stack pointer is incremented by 1 and the contents of that memory location are copied to the high-order register (B, D, H, A) of the operand. The stack pointer register is again incremented by 1. Example: POP H or POP A

Output data from accumulator to a port with 8-bit address

OUT	8-bit port address	The contents of the accumulator are copied into the I/O port specified by the operand. Example: OUT F8H

Input data to accumulator from a port with 8-bit address

IN	8-bit port address	The contents of the input port designated in the operand are read and loaded into the accumulator. Example: IN 8CH

Arithmetic Instructions

Opcode	Operand	Description
Add register or memory to accumulator		
ADD	R M	The contents of the operand (register or memory) are added to the contents of the accumulator and the result is stored in the accumulator. If the operand is a memory location, its location is specified by the contents of the HL registers. All flags are modified to reflect the result of the addition. Example: ADD B or ADD M
Add register to accumulator with carry		
ADC	R M	The contents of the operand (register or memory) and the Carry flag are added to the contents of the accumulator and the result is stored in the accumulator. If the operand is a memory location, its location is specified by the contents of the HL registers. All flags are modified to reflect the result of the addition. Example: ADC B or ADC M
Add immediate to accumulator		
ADI	8-bit data	The 8-bit data (operand) is added to the contents of the accumulator and the result is stored in the accumulator. All flags are modified to reflect the result of the addition. Example: ADI 45H
Add immediate to accumulator with carry		
ACI	8-bit data	The 8-bit data (operand) and the Carry flag are added to the contents of the accumulator and the result is stored in the accumulator. All flags are modified to reflect the result of the addition. Example: ACI 45H
Add register pair to H and L registers		
DAD	Reg. pair	The 16-bit contents of the specified register pair are added to the contents of the HL register and the sum is stored in the HL register. The contents of the source register pair are not altered. If the result is larger than 16 bits, the CY flag is set. No other flags are affected. Example: DAD H

Subtract register or memory from accumulator

SUB	R M	The contents of the operand (register or memory) are subtracted from the contents of the accumulator, and the result is stored in the accumulator. If the operand is a memory location, its location is specified by the contents of the HL registers. All flags are modified to reflect the result of the subtraction. Example: SUB B or SUB M

Subtract source and borrow from accumulator

SBB	R M	The contents of the operand (register or memory) and the Borrow flag are subtracted from the contents of the accumulator and the result is placed in the accumulator. If the operand is a memory location, its location is specified by the contents of the HL registers. All flags are modified to reflect the result of the subtraction. Example: SBB B or SBB M

Subtract immediate from accumulator

SUI	8-bit data	The 8-bit data (operand) is subtracted from the contents of the accumulator and the result is stored in the accumulator. All flags are modified to reflect the result of the subtraction. Example: SUI 45H

Subtract immediate from accumulator with borrow

SBI	8-bit data	The 8-bit data (operand) and the Borrow flag are subtracted from the contents of the accumulator and the result is stored in the accumulator. All flags are modified to reflect the result of the subtracion. Example: SBI 45H

Increment register or memory by 1

INR	R M	The contents of the designated register or memory) are incremented by 1 and the result is stored in the same place. If the operand is a memory location, its location is specified by the contents of the HL registers. Example: INR B or INR M

Increment register pair by 1

INX	R	The contents of the designated register pair are incremented by 1 and the result is stored in the same place. Example: INX H

Decrement register or memory by 1

DCR	R M	The contents of the designated register or memory are decremented by 1 and the result is stored in the same place. If the operand is a memory location, its location is specified by the contents of the HL registers. Example: DCR B or DCR M

Decrement register pair by 1

DCX	R	The contents of the designated register pair are decremented by 1 and the result is stored in the same place. Example: DCX H

Decimal adjust accumulator

DAA	none	The contents of the accumulator are changed from a binary value to two 4-bit binary coded decimal (BCD) digits. This is the only instruction that uses the auxiliary flag to perform the binary to BCD conversion, and the conversion procedure is described below. S, Z, AC, P, CY flags are altered to reflect the results of the operation. If the value of the low-order 4-bits in the accumulator is greater than 9 or if AC flag is set, the instruction adds 6 to the low-order four bits. If the value of the high-order 4-bits in the accumulator is greater than 9 or if the Carry flag is set, the instruction adds 6 to the high-order four bits. Example: DAA

Branching Instructions

Opcode	Operand	Description
Jump unconditionally		
JMP	16-bit address	The program sequence is transferred to the memory location specified by the 16-bit address given in the operand. Example: JMP 2034H or JMP XYZ

Jump conditionally

Operand: 16-bit address

The program sequence is transferred to the memory location specified by the 16-bit address given in the operand based on the specified flag of the PSW as described below.
Example: JZ 2034H or JZ XYZ

Opcode	Description	Flag Status
JC	Jump on Carry	CY = 1
JNC	Jump on no Carry	CY = 0
JP	Jump on positive	S = 0
JM	Jump on minus	S = 1
JZ	Jump on zero	Z = 1
JNZ	Jump on no zero	Z = 0
JPE	Jump on parity even	P = 1
JPO	Jump on parity odd	P = 0

Unconditional subroutine call

CALL 16-bit address

The program sequence is transferred to the memory location specified by the 16-bit address given in the operand. Before the transfer, the address of the next instruction after CALL (the contents of the program counter) is pushed onto the stack. Example: CALL 2034H or CALL XYZ

Call conditionally

Operand: 16-bit address

The program sequence is transferred to the memory location specified by the 16-bit address given in the operand based on the specified flag of the PSW as described below. Before the transfer, the address of the next instruction after the call (the contents of the program counter) is pushed onto the stack. Example: CZ 2034H or CZ XYZ

Opcode	Description	Flag Status
CC	Call on Carry	CY = 1
CNC	Call on no Carry	CY = 0
CP	Call on positive	S = 0
CM	Call on minus	S = 1
CZ	Call on zero	Z = 1
CNZ	Call on no zero	Z = 0
CPE	Call on parity even	P = 1
CPO	Call on parity odd	P = 0

Return from subroutine unconditionally

RET	none	The program sequence is transferred from the subroutine to the calling program. The two bytes from the top of the stack are copied into the program counter, and program execution begins at the new address. Example: RET

Return from subroutine conditionally

Operand: none

The program sequence is transferred from the subroutine to the calling program based on the specified flag of the PSW as described below. The two bytes from the top of the stack are copied into the program counter, and program execution begins at the new address.
Example: RZ

Opcode	Description	Flag Status
RC	Return on Carry	CY = 1
RNC	Return on no Carry	CY = 0
RP	Return on positive	S = 0
RM	Return on minus	S = 1
RZ	Return on zero	Z = 1
RNZ	Return on no zero	Z = 0
RPE	Return on parity even	P = 1
RPO	Return on parity odd	P = 0

Load program counter with HL contents

PCHL none

The contents of registers H and L are copied into the program counter. The contents of H are placed as the high-order byte and the contents of L as the low-order byte.
Example: PCHL

Restart

RST 0-7

The RST instruction is equivalent to a 1-byte call instruction to one of eight memory locations depending upon the number. The instructions are generally used in conjunction with interrupts and inserted using external hardware. However these can be used as software instructions in a program to transfer program execution to one of the eight locations. The addresses are:

Instruction	Restart Address
RST 0	0000H
RST 1	0008H
RST 2	0010H
RST 3	0018H
RST 4	0020H
RST 5	0028H
RST 6	0030H
RST 7	0038H

The 8085 has four additional interrupts and these interrupts generate RST instructions internally and thus do not require any external hardware. These instructions and their Restart addresses are:

Interrupt	Restart Address
TRAP	0024H
RST 5.5	002CH
RST 6.5	0034H
RST 7.5	003CH

Logical Instructions

Opcode	Operand	Description
Compare register or memory with accumulator		

CMP	R
	M
		The contents of the operand (register or memory) are compared with the contents of the accumulator. Both contents are preserved . The result of the comparison is shown by setting the flags of the PSW as follows:
		if (A) < (reg/mem): carry flag is set if
		(A) = (reg/mem): zero flag is set
		if (A) > (reg/mem): carry and zero flags are reset
		Example: CMP B or CMP M

Compare immediate with accumulator		

CPI	8-bit data	The second byte (8-bit data) is compared with the contents of the accumulator. The values being compared remain unchanged. The result of the comparison is shown by setting the flags of the PSW as follows:
		if (A) < data: carry flag is set if
		(A) = data: zero flag is set
		if (A) > data: carry and zero flags are reset Example:
		CPI 89H

Logical AND register or memory with accumulator		
ANA	R M	The contents of the accumulator are logically ANDed with the contents of the operand (register or memory), and the result is placed in the accumulator. If the operand is a memory location, its address is specified by the contents of HL registers. S, Z, P are modified to reflect the result of the operation. CY is reset. AC is set. Example: ANA B or ANA M
Logical AND immediate with accumulator		
ANI	8-bit data	The contents of the accumulator are logically ANDed with the 8-bit data (operand) and the result is placed in the accumulator. S, Z, P are modified to reflect the result of the operation. CY is reset. AC is set. Example: ANI 86H

Exclusive OR register or memory with accumulator

XRA	R M	The contents of the accumulator are Exclusive ORed with the contents of the operand (register or memory), and the result is placed in the accumulator. If the operand is a memory location, its address is specified by the contents of HL registers. S, Z, P are modified to reflect the result of the operation. CY and AC are reset. Example: XRA B or XRA M

Exclusive OR immediate with accumulator

XRI	8-bit data	The contents of the accumulator are Exclusive ORed with the 8-bit data (operand) and the result is placed in the accumulator. S, Z, P are modified to reflect the result of the operation. CY and AC are reset. Example: XRI 86H

Logical OR register or memory with accumulaotr

ORA	R M	The contents of the accumulator are logically ORed with the contents of the operand (register or memory), and the result is placed in the accumulator. If the operand is a memory location, its address is specified by the contents of HL registers. S, Z, P are modified to reflect the result of the operation. CY and AC are reset. Example: ORA B or ORA M

Logical OR immediate with accumulator

ORI	8-bit data	The contents of the accumulator are logically ORed with the 8-bit data (operand) and the result is placed in the accumulator. S, Z, P are modified to reflect the result of the operation. CY and AC are reset. Example: ORI 86H

Rotate accumulator left

RLC	none	Each binary bit of the accumulator is rotated left by one position. Bit D_7 is placed in the position of D_0 as well as in the Carry flag. CY is modified according to bit D_7. S, Z, P, AC are not affected. Example: RLC

Rotate accumulator right

RRC	none	Each binary bit of the accumulator is rotated right by one position. Bit D_0 is placed in the position of D_7 as well as in the Carry flag. CY is modified according to bit D_0. S, Z, P, AC are not affected. Example: RRC

Rotate accumulator left through carry

RAL	none	Each binary bit of the accumulator is rotated left by one position through the Carry flag. Bit D_7 is placed in the Carry flag, and the Carry flag is placed in the least significant position D_0. CY is modified according to bit D_7. S, Z, P, AC are not affected. Example: RAL

Rotate accumulator right through carry

RAR	none	Each binary bit of the accumulator is rotated right by one position through the Carry flag. Bit D_0 is placed in the Carry flag, and the Carry flag is placed in the most significant position D_7. CY is modified according to bit D_0. S, Z, P, AC are not affected. Example: RAR

Complement accumulator

CMA	none	The contents of the accumulator are complemented. No flags are affected. Example: CMA

Complement carry

CMC	none	The Carry flag is complemented. No other flags are affected. Example: CMC

Set Carry

STC	none	The Carry flag is set to 1. No other flags are affected. Example: STC

Control Instructions

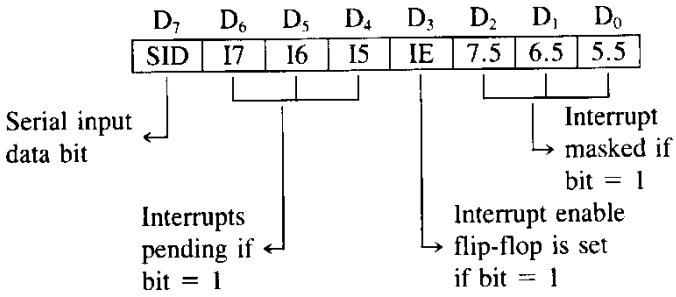
Opcode	Operand	Description
No operation		
NOP	none	No operation is performed. The instruction is fetched and decoded. However no operation is executed. Example: NOP
Halt and enter wait state		
HLT	none	The CPU finishes executing the current instruction and halts any further execution. An interrupt or reset is necessary to exit from the halt state. Example: HLT
Disable interrupts		
DI	none	The interrupt enable flip-flop is reset and all the interrupts except the TRAP are disabled. No flags are affected. Example: DI
Enable interrupts		
EI	none	The interrupt enable flip-flop is set and all interrupts are enabled. No flags are affected. After a system reset or the acknowledgement of an interrupt, the interrupt enable flip-flop is reset, thus disabling the interrupts. This instruction is necessary to reenable the interrupts (except TRAP). Example: EI

Read interrupt mask

RIM none

This is a multipurpose instruction used to read the status of interrupts 7.5, 6.5, 5.5 and read serial data input bit. The instruction loads eight bits in the accumulator with the following interpretations.

Example: RIM



Set interrupt mask

SIM none

This is a multipurpose instruction and used to implement the 8085 interrupts 7.5, 6.5, 5.5, and serial data output. The instruction interprets the accumulator contents as follows.

Example: SIM

Chapter 5

Memories and I/O interfacing

The programs and data that are executed by the microprocessor have to be stored in ROM/EPROM and RAM, which are basically semiconductor memory chips. The programs and data that are stored in ROM/EPROM are not erased even when power supply to the chip is removed. Hence, they are called non-volatile memory. They can be used to store permanent programs.

In a RAM, stored programs and data are erased when the power supply to the chip is removed. Hence, RAM is called volatile memory. RAM can be used to store programs and data that include, programs written during software development for a microprocessor based system, program written when one is learning assembly language programming and data enter while testing these programs.

Input and output devices, which are interfaced with 8085, are essential in any microprocessor based system. They can be interfaced using two schemes: I/O mapped I/O and memory-mapped I/O. In the I/O mapped I/O scheme, the I/O devices are treated differently from memory. In the memory-mapped I/O scheme, each I/O device is assumed to be a memory location.

Interfacing Memory Chips With 8085

8085 has 16 address lines (A0 - A15), hence a maximum of 64 KB (= 2^{16} bytes) of memory locations can be interfaced with it. The memory address space of the 8085 takes values from 0000H to FFFFH.

The 8085 initiates set of signals such as IO/$\overline{M}$, $\overline{RD}$ and $\overline{WR}$ when it wants to read from and write into memory. Similarly, each memory chip has signals such as $\overline{CE}$ or $\overline{CS}$ (chip enable or chip select), $\overline{OE}$ or $\overline{RD}$ (output enable or read) and $\overline{WE}$ or $\overline{WR}$ (write enable or write) associated with it.

Generation of Control Signals for Memory:

When the 8085 wants to read from and write into memory, it activates IO/$\overline{M}$, $\overline{RD}$ and $\overline{WR}$ signals as shown in Table 8.

Table 8 Status of IO/$\overline{M}$, $\overline{RD}$ and $\overline{WR}$ signals during memory read and write operations

IO/$\overline{M}$	$\overline{RD}$	$\overline{WR}$	Operation
0	0	1	8085 reads data from memory
0	1	0	8085 writes data into memory

Using $IO/\overline{M}$, $\overline{RD}$ and $\overline{WR}$ signals, two control signals $\overline{MEMR}$ (memory read) and $\overline{MEMW}$ (memory write) are generated. Fig. shows the circuit used to generate these signals.

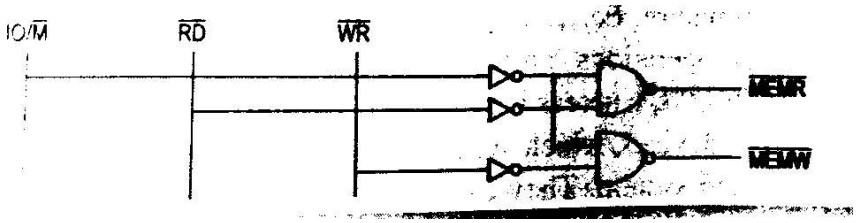


Fig. Circuit used to generate $\overline{MEMR}$ and $\overline{MEMW}$ signals

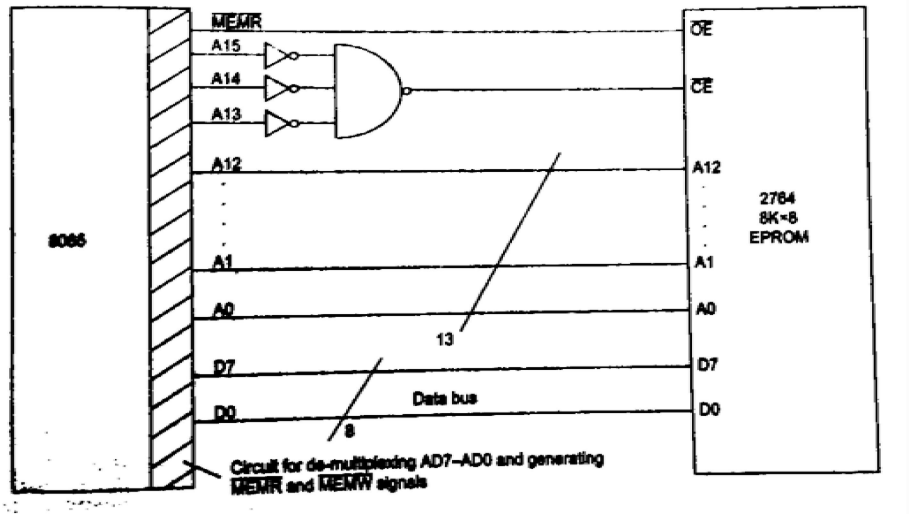
Ex: Interface an IC 2764 with 8085 using NAND gate address decoder such that the address range allocated to the chip is 0000H – 1FFFH.

Specification of IC 2764:

- 8 KB (8 x 2^{10} byte) EPROM chip
- 13 address lines (2^{13} bytes = 8 KB)

Interfacing:

- 13 address lines of IC are connected to the corresponding address lines of 8085.
- Remaining address lines of 8085 are connected to address decoder formed using logic gates, the output of which is connected to the CE pin of IC.
- Address range allocated to the chip is shown in Table 9.
- Chip is enabled whenever the 8085 places an address allocated to EPROM chip in the address bus. This is shown in Fig..



Interfacing IC 2764 with the 8085

Address allocated to IC 2764

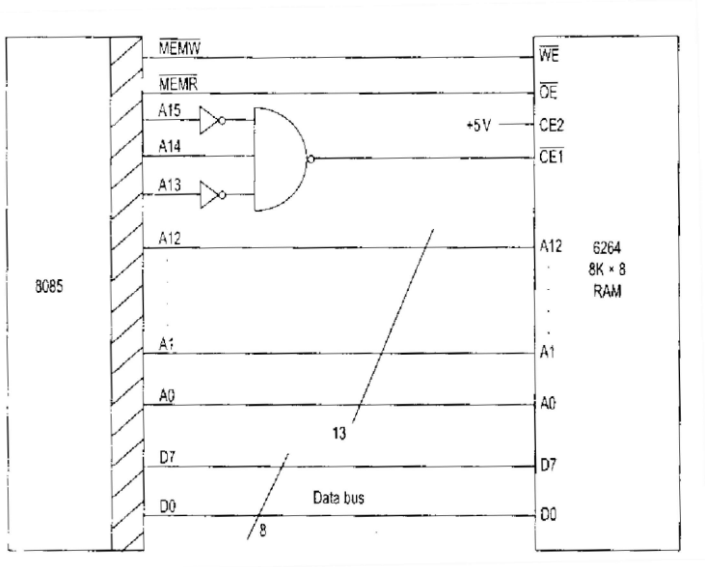
A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0	Address
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0000H
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0001H
.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	0	1FFEH
0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1FFFH

Ex: Interface a 6264 IC (8K x 8 RAM) with the 8085 using NAND gate decoder such that the starting address assigned to the chip is 4000H.

Specification of IC 6264:

- 8K x 8 RAM
- 8 KB = 2^{13} bytes
- 13 address lines

The ending address of the chip is 5FFFH (since 4000H + 1FFFH = 5FFFH). When the address 4000H to 5FFFH are written in binary form, the values in the lines A15, A14, A13 are 0, 1 and 0 respectively. The NAND gate is designed such that when the lines A15 and A13 carry 0 and A14 carries 1, the output of the NAND gate is 0. The NAND gate output is in turn connected to the $\overline{CE1}$ pin of the RAM chip. A NAND output of 0 selects the RAM chip for read or write operation, since CE2 is already 1 because of its connection to +5V. Fig. shows the interfacing of IC 6264 with the 8085.



Interfacing 6264 IC with the 8085

Ex: Interface two 6116 ICs with the 8085 using 74LS138 decoder such that the starting addresses assigned to them are 8000H and 9000H, respectively.

Specification of IC 6116:

- 2 K x 8 RAM
- 2 KB = 2^{11} bytes
- 11 address lines

6116 has 11 address lines and since 2 KB, therefore ending addresses of 6116 chip 1 is and chip 2 are 87FFH and 97FFH, respectively. Table 10 shows the address range of the two chips.

Address range for IC 6116

A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0	Address
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8000H
.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	87FFH (RAM chip 1)
1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	9000H
.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
1	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	97FFH (RAM chip 2)

Interfacing:

- Fig. shows the interfacing.
- A0 – A10 lines of 8085 are connected to 11 address lines of the RAM chips.
- Three address lines of 8085 having specific value for a particular RAM are connected to the three select inputs (C, B and A) of 74LS138 decoder.
- Table 10 shows that A13=A12=A11=0 for the address assigned to RAM 1 and A13=0, A12=1 and A11=0 for the address assigned to RAM 2.
- Remaining lines of 8085 which are constant for the address range assigned to the two RAM are connected to the enable inputs of decoder.
- When 8085 places any address between 8000H and 87FFH in the address bus, the select inputs C, B and A of the decoder are all 0. The Y0 output of the decoder is also

0, selecting RAM 1.

- When 8085 places any address between 9000H and 97FFH in the address bus, the select inputs C, B and A of the decoder are 0, 1 and 0. The Y2 output of the decoder is also 0, selecting RAM 2.

Peripheral Mapped I/O Interfacing

In this method, the I/O devices are treated differently from memory chips. The control signals I/O read ($\overline{IOR}$) and I/O write ($\overline{IOW}$), which are derived from the IO/$\overline{M}$, $\overline{RD}$ and $\overline{WR}$ signals of the 8085, are used to activate input and output devices, respectively. Generation of these control signals is shown in Fig. and Table shows the status of IO/$\overline{M}$, $\overline{RD}$ and $\overline{WR}$ signals during I/O read and I/O write operation.

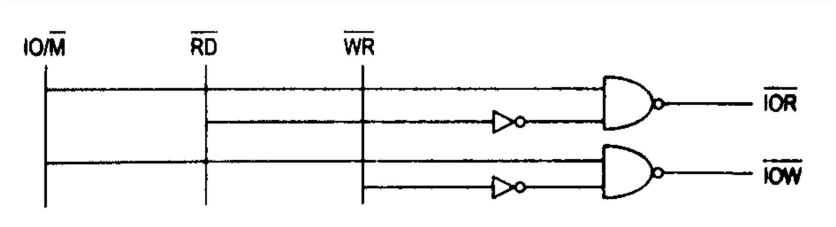


Fig. Generation of $\overline{IOR}$ and $\overline{IOW}$ signals

IN instruction is used to access input device and OUT instruction is used to access output device. Each I/O device is identified by a unique 8-bit address assigned to it. Since the control signals used to access input and output devices are different, and all I/O device use 8-bit address, a maximum of 256 (2^8) input devices and 256 output devices can be interfaced with 8085.

Status of $\overline{IOR}$ and $\overline{IOW}$ signals in 8085.

IO/$\overline{M}$	$\overline{RD}$	$\overline{WR}$	$\overline{IOR}$	$\overline{IOW}$	Operation
1	0	1	0	1	I/O read operation
1	1	0	1	0	I/O write operation
0	X	X	1	1	Memory read or write operation

Ex: Interface an 8-bit DIP switch with the 8085 such that the address assigned to the DIP switch if F0H.

IN instruction is used to get data from DIP switch and store it in accumulator. Steps involved

in the execution of this instruction are:

i. Address F0H is placed in the lines A0 – A7 and a copy of it in lines A8 – A15.
ii. The $\overline{IOR}$ signal is activated ($\overline{IOR} = 0$), which makes the selected input device to place its data in the data bus.
iii. The data in the data bus is read and store in the accumulator.

A7	A6	A5	A4	A3	A2	A1	A0	
1	1	1	1	0	0	0	0	= F0H

A0 – A7 lines are connected to a NAND gate decoder such that the output of NAND gate is 0. The output of NAND gate is ORed with the $\overline{IOR}$ signal and the output of OR gate is connected to $1\overline{G}$ and $2\overline{G}$ of the 74LS244. When 74LS244 is enabled, data from the DIP switch is placed on the data bus of the 8085. The 8085 read data and store in the accumulator. Thus data from DIP switch is transferred to the accumulator.

Memory Mapped I/O Interfacing

In memory-mapped I/O, each input or output device is treated as if it is a memory location. The $\overline{MEMR}$ and $\overline{MEMW}$ control signals are used to activate the devices. Each input or output device is identified by unique 16-bit address, similar to 16-bit address assigned to memory location. All memory related instruction like LDA 2000H, LDAX B, MOV A, M can be used. Since the I/O devices use some of the memory address space of 8085, the maximum memory capacity is lesser than 64 KB in this method.

Ex: Interface an 8-bit DIP switch with the 8085 using logic gates such that the address assigned to it is F0F0H.

Since a 16-bit address has to be assigned to a DIP switch, the memory-mapped I/O technique must be used. Using LDA F0F0H instruction, the data from the 8-bit DIP switch can be transferred to the accumulator. The steps involved are:

i. The address F0F0H is placed in the address bus A0 – A15.
ii. The $\overline{MEMR}$ signal is made low for some time.
iii. The data in the data bus is read and stored in the accumulator.

Fig. shows the interfacing diagram.

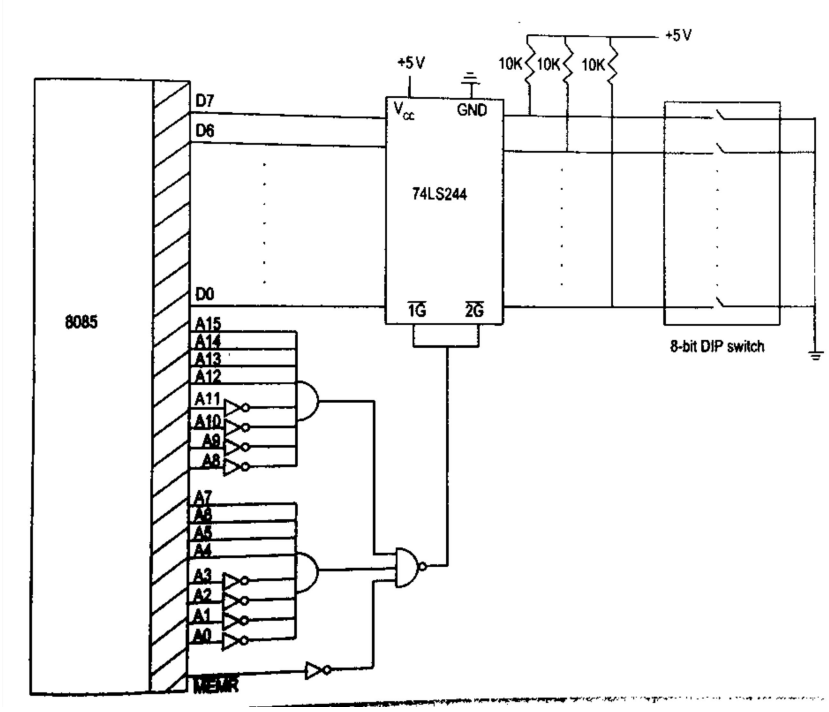


Fig. Interfacing 8-bit DIP switch with 8085

When 8085 executes the instruction LDA F0F0H, it places the address F0F0H in the address lines A0 – A15 as:

A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0	
1	1	1	1	0	0	0	0	1	1	1	1	0	0	0	0	= F0F0H

The address lines are connected to AND gates. The output of these gates along with $\overline{\text{MEMR}}$ signal are connected to a NAND gate, so that when the address F0F0H is placed in the address bus and $\overline{\text{MEMR}} = 0$ its output becomes 0, thereby enabling the buffer 74LS244. The data from the DIP switch is placed in the 8085 data bus. The 8085 reads the data from the data bus and stores it in the accumulator.

Mode of Transfer:

The binary information that is received from an external device is usually stored in the memory unit. The information that is transferred from the CPU to the external device is originated from the memory unit. CPU merely processes the information but the source and target is always the memory unit. Data transfer between CPU and the I/O devices may be done in different modes.

Data transfer to and from the peripherals may be done in any of the three possible ways

1. Programmed I/O.
2. Interrupt- initiated I/O.
3. Direct Memory Access (DMA).

Now let's discuss each mode one by one.

1. **Programmed I/O:** It is due to the result of the I/O instructions that are written in the computer program. Each data item transfer is initiated by an instruction in the program. Usually the transfer is from a CPU register and memory. In this case it requires constant monitoring by the CPU of the peripheral devices.

   Example of Programmed I/O: In this case, the I/O device does not have direct access to the memory unit. A transfer from I/O device to memory requires the execution of several instructions by the CPU, including an input instruction to transfer the data from device to the CPU and store instruction to transfer the data from CPU to memory. In programmed I/O, the CPU stays in the program loop until the I/O unit indicates that it is ready for data transfer. This is a time consuming process since it needlessly keeps the CPU busy. This situation can be avoided by using an interrupt facility. This is discussed below.

2. **Interrupt- initiated I/O:** Since in the above case we saw the CPU is kept busy unnecessarily. This situation can very well be avoided by using an interrupt driven method for data transfer. By using interrupt facility and special commands to inform the interface to issue an interrupt request signal whenever data is available from any device. In the meantime the CPU can proceed for any other program execution. The interface meanwhile keeps monitoring the device. Whenever it is determined that the device is ready for data transfer it initiates an interrupt request signal to the computer. Upon detection of an external interrupt signal the CPU stops momentarily the task that it was already performing, branches to the service program to process the I/O transfer, and then return to the task it was originally performing.

3. **Direct Memory Access**: The data transfer between a fast storage media such as magnetic disk and memory unit is limited by the speed of the CPU. Thus we can allow the peripherals directly communicate with each other using the memory buses, removing the intervention of the CPU. This type of data transfer technique is known as DMA or direct memory access. During DMA the CPU is idle and it

has no control over the memory buses. The DMA controller takes over the buses to manage the transfer directly between the I/O devices and the memory unit.

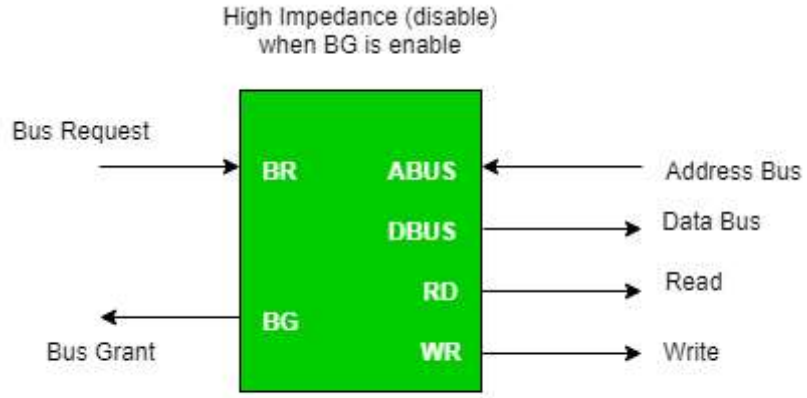


Figure - CPU Bus Signals for DMA Transfer

Bus Request: It is used by the DMA controller to request the CPU to relinquish the control of the buses.

Bus Grant: It is activated by the CPU to Inform the external DMA controller that the buses are in high impedance state and the requesting DMA can take control of the buses. Once the DMA has taken the control of the buses it transfers the data. This transfer can take place in many ways.

Example it can transfer using

Bus Transfer: In which a block sequence consisting of memory words is transferred in a continuous burst where the DMA controller is the master of the memory buses. This mode is needed for fast devices like magnetic disks.

Cyclic Stealing: In this DMA controller transfers one word at a time after which it must return the control of the buses to the CPU. The CPU merely delays its operation for one memory cycle to allow the direct memory I/O transfer to “steal” one memory cycle.

Chapter 6

Interrupt

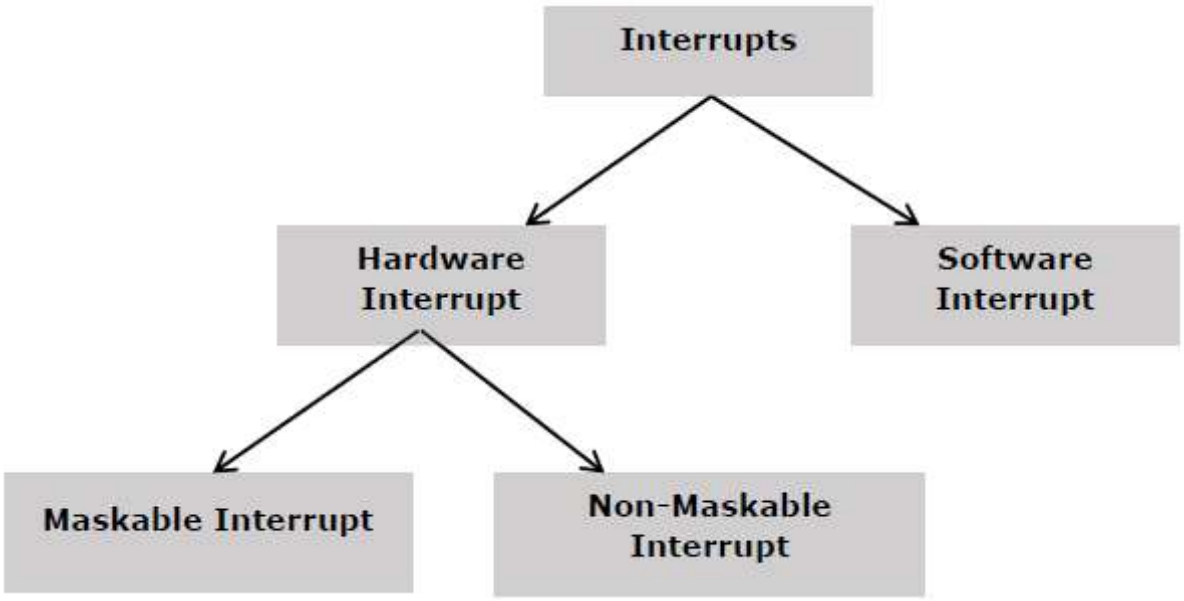
In system programming, an interrupt is a signal to the processor emitted by hardware or software indicating an event that needs immediate attention. An interrupt alerts the processor to a high-priority condition requiring the interruption of the current code the processor is executing. The processor responds by suspending its current activities, saving its state, and executing a function called an interrupt handler (or an interrupt service routine, ISR) to deal with the event. This interruption is temporary, and, after the interrupt handler finishes, the processor resumes normal activities.

There are two types of interrupts: hardware interrupts and software interrupts.
Hardware interrupts are used by devices to communicate that they require attention from the operating system. Internally, hardware interrupts are implemented using electronic alerting signals that are sent to the processor from an external device, which is either a part of the computer itself, such as a disk controller, or an external peripheral. For example, pressing a key on the keyboard or moving the mouse triggers hardware interrupts that cause the processor to read the keystroke or mouse position. Unlike the software type (described below), hardware interrupts are asynchronous and can occur in the middle of instruction execution, requiring additional care in programming. The act of initiating a hardware interrupt is referred to as an interrupt request (IRQ).
Software interrupt is caused either by an exceptional condition in the processor itself, or a special instruction in the instruction set which causes an interrupt when it is executed. The former is often called a trap or exception and is used for errors or events occurring during program executions that are exceptional enough that they cannot be handled within the program itself. For example, a divide-by-zero exception will be thrown if the processor's arithmetic logic unit is commanded to divide a number by zero as this instruction is an error and impossible. The operating system will catch this exception, and can choose to abort the instruction. Software interrupt instructions can function similarly to subroutine calls and are used for a variety of purposes, such as to request services from device drivers, like interrupts sent to and from a disk controller to request reading or writing of data to and from the disk.
Each interrupt has its own interrupt handler. The number of hardware interrupts is limited by the number of interrupt request (IRQ) lines to the processor, but there may be hundreds of different software interrupts. Interrupts are a commonly used technique for computer multitasking, especially in real-time computing. Such a system is said to be interrupt-driven.[3]
Interrupts are similar to signals, the difference being that signals are used for IPC, mediated by the kernel (possibly via system calls) and handled by processes, while interrupts are mediated by the processor and handled by the kernel. The kernel may pass an interrupt as a signal to the process that caused it (typical examples are SIGSEGV, SIGBUS, SIGILL and SIGFPE).

Interrupt

Interrupt is the method of creating a temporary halt during program execution and allows peripheral devices to access the microprocessor. The microprocessor responds to that interrupt with an **ISR** (Interrupt Service Routine), which is a short program to instruct the microprocessor on how to handle the interrupt.

The following image shows the types of interrupts we have in a 8086 microprocessor −



Interrupt

Hardware Interrupts

Hardware interrupt is caused by any peripheral device by sending a signal through a specified pin to the microprocessor.

The 8086 has two hardware interrupt pins, i.e. NMI and INTR. NMI is a non-mask able interrupt and INTR is a mask-able interrupt having lower priority. One more interrupt pin associated is INTA called interrupt acknowledge.

NMI: It is a single non-mask-able interrupt pin (NMI) having higher priority than the mask-able interrupt request pin (INTR) and it is of type 2 interrupt.

When this interrupt is activated, these actions take place −

- Completes the current instruction that is in progress.
- Pushes the Flag register values on to the stack.
- Pushes the CS (code segment) value and IP (instruction pointer) value of the return address on to the stack.
- IP is loaded from the contents of the word location 00008H.
- CS is loaded from the contents of the next word location 0000AH.
- Interrupt flag and trap flag are reset to 0.

INTR: The INTR is a mask-able interrupt because the microprocessor will be interrupted only if interrupts are enabled using set interrupt flag instruction. It should not be enabled using clear interrupt Flag instruction.

The INTR interrupt is activated by an I/O port. If the interrupt is enabled and NMI is disabled, then the microprocessor first completes the current execution and sends '0' on INTA pin twice. The first '0' means INTA informs the external device to get ready and during the second '0' the microprocessor receives the 8 bit, say X, from the programmable interrupt controller.

These actions are taken by the microprocessor −

- First completes the current instruction.
- Activates INTA output and receives the interrupt type, say X.
- Flag register value; CS value of the return address and IP value of the return address are pushed on to the stack.
- IP value is loaded from the contents of word location X × 4
- CS is loaded from the contents of the next word location.
- Interrupt flag and trap flag is reset to 0

Software Interrupts

Some instructions are inserted at the desired position into the program to create interrupts. These interrupt instructions can be used to test the working of various interrupt handlers. It includes –

INT- Interrupt instruction with type number

It is 2-byte instruction. First byte provides the op-code and the second byte provides the interrupt type number. There are 256 interrupt types under this group.

Its execution includes the following steps −

- Flag register value is pushed on to the stack.
- CS value of the return address and IP value of the return address are pushed on to the stack.
- IP is loaded from the contents of the word location 'type number'× 4.
- CS is loaded from the contents of the next word location.
- Interrupt Flag and Trap Flag are reset to 0

The starting address for type0 interrupt is 000000H, for type1 interrupt is 00004H similarly for type2 is 00008H and ……so on. The first five pointers are dedicated interrupt pointers. i.e. -

- **TYPE 0** interrupt represents division by zero situation.
- **TYPE 1** interrupt represents single-step execution during the debugging of a program.
- **TYPE 2** interrupt represents non-mask-able NMI interrupt.
- **TYPE 3** interrupt represents break-point interrupt.
- **TYPE 4** interrupt represents overflow interrupt.

The interrupts from Type 5 to Type 31 are reserved for other advanced microprocessors, and interrupts from 32 to Type 255 are available for hardware and software interrupts.

INT 3-Break Point Interrupt Instruction

It is a 1-byte instruction having op-code is CCH. These instructions are inserted into the program so that when the processor reaches there, then it stops the normal execution of program and follows the break-point procedure.

Its execution includes the following steps −

- Flag register value is pushed on to the stack.
- CS value of the return address and IP value of the return address are pushed on to the stack.
- IP is loaded from the contents of the word location 3×4 = 0000CH
- CS is loaded from the contents of the next word location.
- Interrupt Flag and Trap Flag are reset to 0.

INTO - Interrupt on overflow instruction

It is a 1-byte instruction and their mnemonic **INTO**. The op-code for this instruction is CEH. As the name suggests it is a conditional interrupt instruction, i.e. it is active only when the overflow flag is set to 1 and branches to the interrupt handler whose interrupt type number is 4. If the overflow flag is reset then, the execution continues to the next instruction.

Its execution includes the following steps −

- Flag register values are pushed on to the stack.
- CS value of the return address and IP value of the return address are pushed on to the stack.
- IP is loaded from the contents of word location 4×4 = 00010H
- CS is loaded from the contents of the next word location.
- Interrupt flag and Trap flag are reset to 0

Non-Mask-Able Interrupts

A non-mask-able interrupt (NMI) is a type of hardware interrupt (or signal to the processor) that prioritizes a certain thread or process. Unlike other types of interrupts, the non-mask-able interrupt cannot be ignored through the use of interrupt masking techniques.

Mask-Able interrupt

An Interrupt that can be disabled or ignored by the instructions of CPU is called as mask-able Interrupt.

Eg: RST6.5, RST7.5, RST5.5 OF 8085 is mask-able Interrupts.

Chapter 7

Data transfer Techniques

Programmed input/output (**PIO**) is a method of transferring data between the CPU and a peripheral, such as a network adapter or an ATA storage device.
For programmed I/O, the software that is running on the CPU uses instructions to perform data transfers to or from an I/O device. This is in contrast to Direct Memory Access (DMA) transfers. The term *Programmed I/O* can refer to either MMIO or PMIO. Port-mapped I/O (PMIO) refers to a special address space outside of normal memory that is accessed with instructions such as IN and OUT. Memory-mapped I/O[1] (MMIO) refers to I/O devices being allocated addresses inside the normal Von Neumann address space that is primarily used for program and data. Such I/O is done using instructions such as LOAD and STORE. PMIO was very useful for early microprocessors with small address spaces, since the valuable resource was not consumed by the I/O devices.

The best known example of a PC device that uses programmed I/O is the ATA interface; however, this interface can also be operated in any of several DMA modes. Many older devices in a PC also use PIO, including legacy serial ports, legacy parallel ports when not in ECP mode, the PS/2 keyboard and mouse ports, legacy MIDI and joystick ports, the interval timer, and older network interfaces.

Synchronous data transmission

Synchronous transmission is transmission of signals in a fixed interval based on a predefined clocking signal and is meant for constant and reliable transmission of time-sensitive data such as VoIP and audio/video streaming.
This method of transmission is used when large amounts of data need to be transferred quickly since data is transferred in large blocks instead of individual characters. The data blocks are spaced and grouped in regular intervals and preceded by synchronous characters that a remote device decode and use to synchronize the connection between the end points.
After synchronization is complete, the transmission can begin.

Asynchronous data transfer

The transmission of asynchronous data is not prompted by a clock signal when sending the data to the receiver, unlike in synchronous methods, where sending data is measured against a time reference. Compared to synchronous transmission, asynchronous communication has a few advantages:

- It is more flexible and devices can exchange information at their own pace. Individual data characters can complete themselves so that even if one packet is corrupted, its predecessors and successors will not be affected.
- It does not require complex processes by the receiving device. This means that an inconsistency in the transmission of data does not result in a big crisis, since the device can keep up with the data stream. This also makes asynchronous

transfers suitable for applications where character data is generated in an irregular manner.

There are also some disadvantages of using asynchronous data for transmission:

- The success of these transmissions depends on the start bits and their recognition. This can be easily susceptible to line interference, causing these bits to be corrupted or distorted.
- A large portion of the transmitted data is used for control and identification bits for headers and thus carries no useful information related to the transmitted data. This invariably means that more data packets need to be sent.

Interrupt driven I/O

Interrupt driven I/O is an alternative scheme dealing with I/O. Interrupt I/O is a way of controlling input/output activity whereby a peripheral or terminal that needs to make or receive a data transfer sends a signal. This will cause a program interrupt to be set. At a time appropriate to the priority level of the I/O interrupt. Relative to the total interrupt system, the processors enter an interrupt service routine. The function of the routine will depend upon the system of interrupt levels and priorities that is implemented in the processor. The interrupt technique requires more complex hardware and software, but makes far more efficient use of the computer's time and capacities

For **input**, the device interrupts the CPU when new data has arrived and is ready to be retrieved by the system processor. The actual actions to perform depend on whether the device uses I/O ports or memory mapping.

For **output**, the device delivers an interrupt either when it is ready to accept new data or to acknowledge a successful data transfer. Memory-mapped and DMA-capable devices usually generate interrupts to tell the system they are done with the buffer.

Here the CPU works on its given tasks continuously. When an input is available, such as when someone types a key on the keyboard, then the CPU is interrupted from its work to take care of the input data. The CPU can work continuously on a task without checking the input devices, allowing the devices themselves to interrupt it as necessary.

Direct Memory Access

Direct memory access (**DMA**) is a feature of computer systems that allows certain hardware subsystems to access main system memory (Random-access memory), independent of the central processing unit (CPU).

Without DMA, when the CPU is using programmed input/output, it is typically fully occupied for the entire duration of the read or write operation, and is thus unavailable to perform other work. With DMA, the CPU first initiates the transfer, then it does other operations while the transfer is in progress, and it finally receives an interrupt from the DMA controller when the operation is done. This feature is useful at any time that the CPU cannot keep up with the rate of data transfer, or when the CPU needs to perform work while waiting for a relatively slow I/O data transfer. Many hardware systems use DMA, including disk drivecontrollers, graphics cards, network cards and sound cards. DMA is also used for intra-chip data transfer in multi-core

processors. Computers that have DMA channels can transfer data to and from devices with much less CPU overhead than computers without DMA channels. Similarly, a processing element inside a mufti-core processor can transfer data to and from its local memory without occupying its processor time, allowing computation and data transfer to proceed in parallel.
DMA can also be used for "memory to memory" copying or moving of data within memory. DMA can offload expensive memory operations, such as large copies or scatter-gather operations, from the CPU to a dedicated DMA engine. An implementation example is the I/O Acceleration Technology.

Serial I/O

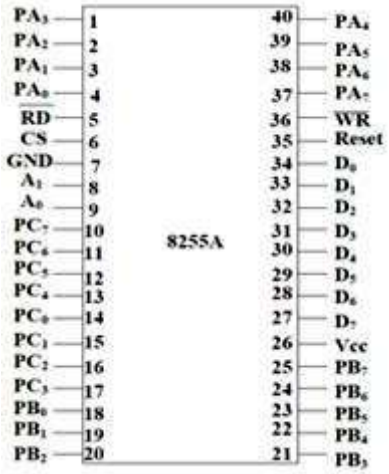
Serial input/output (SIO) A method of communicating data between devices, typically a computer and its peripherals, the individual data bits being sent sequentially. Serial communication may be *asynchronous*, where the data characters include start and stop bits to delimit the data, or *synchronous*, where such additional bits are omitted and the delimiting of the data depends purely on timing. Asynchronous serial communication is more flexible whereas synchronous serial communication makes better use of the available bandwidth. Asynchronous methods are generally used with dial-up modems or for general connection of simple serial peripherals. Synchronous methods are usually to be found where leased lines or proprietary interfaces are used.

Until recently SIO was a relatively slow mode, but advances in silicon technology means that very fast SIO (hundreds of megabits per second) now make SIO a very cost-effective alternative to parallel interfaces. See USB, Firewire, serial IDE.

Chapter 8
Peripheral devices

8255 Pin Diagram

Let us first take a look at the pin diagram of Intel 8255A −



pin diagram 8255 A

Now let us discuss the functional description of the pins in 8255A.

Data Bus Buffer

It is a tri-state 8-bit buffer, which is used to interface the microprocessor to the system data bus. Data is transmitted or received by the buffer as per the instructions by the CPU. Control words and status information is also transferred using this bus.

Read/Write Control Logic

This block is responsible for controlling the internal/external transfer of data/control/status word. It accepts the input from the CPU address and control buses, and in turn issues command to both the control groups.

CS

It stands for Chip Select. A LOW on this input selects the chip and enables the communication between the 8255A and the CPU. It is connected to the decoded address, and A_0 & A_1 are connected to the microprocessor address lines.

WR

It stands for write. This control signal enables the write operation. When this signal goes low, the microprocessor writes into a selected I/O port or control register.

RESET

This is an active high signal. It clears the control register and sets all ports in the input mode.

RD

It stands for Read. This control signal enables the Read operation. When the signal is low, the microprocessor reads the data from the selected I/O port of the 8255.

8257

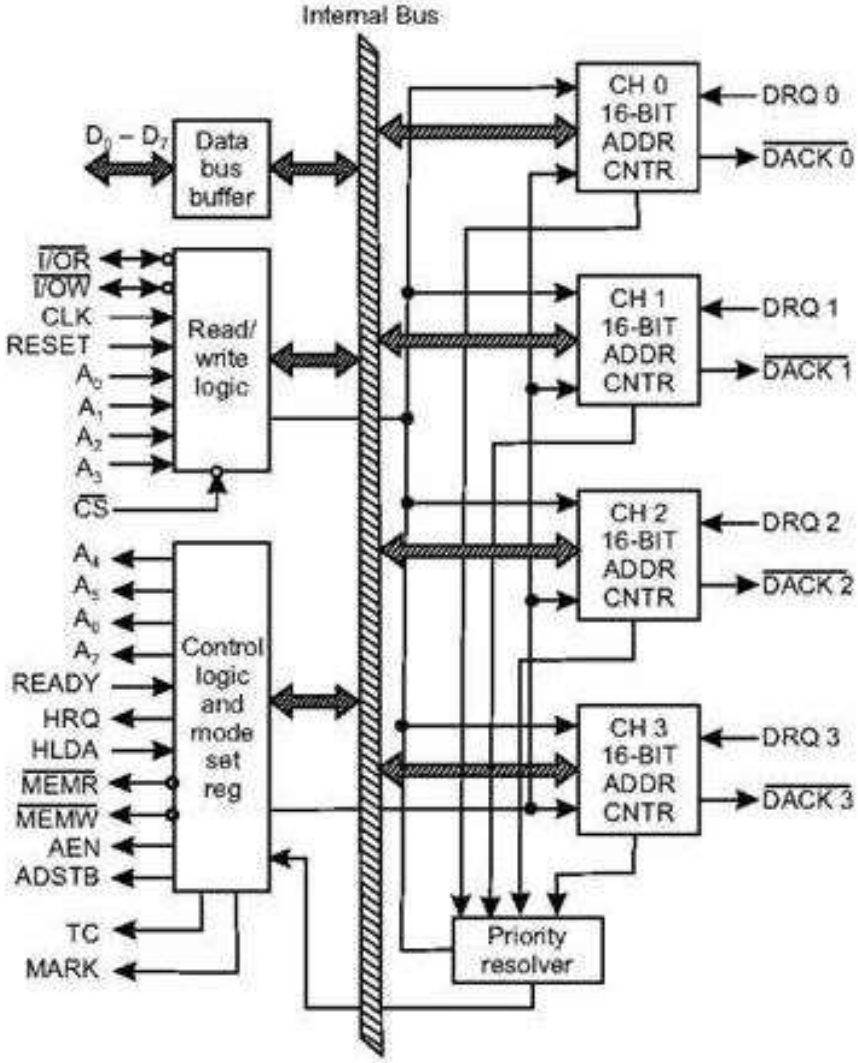
Here is a list of some of the prominent features of 8257 −

- It has four channels which can be used over four I/O devices.

- Each channel has 16-bit address and 14-bit counter.
- Each channel can transfer data up to 64kb.
- Each channel can be programmed independently.
- Each channel can perform read transfer, write transfer and verify transfer operations.
- It generates MARK signal to the peripheral device that 128 bytes have been transferred.
- It requires a single phase clock.
- Its frequency ranges from 250Hz to 3MHz.
- It operates in 2 modes, i.e., **Master mode** and **Slave mode**.

8257 Architecture

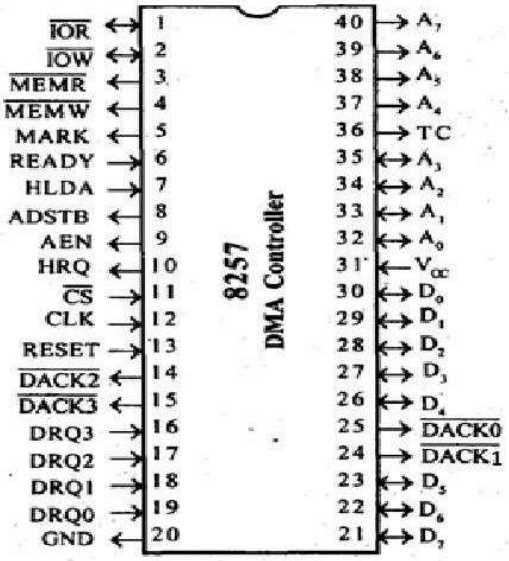
The following image shows the architecture of 8257 −



Pin diagram of 8257

8257 Pin Description

The following image shows the pin diagram of a 8257 DMA controller −



8257 pin description

DRQ_0−DRQ3

These are the four individual channel DMA request inputs, which are used by the peripheral devices for using DMA services. When the fixed priority mode is selected, then DRQ_0 has the highest priority and DRQ_3 has the lowest priority among them.

$DACK_0$ − $DACK_3$

These are the active-low DMA acknowledge lines, which updates the requesting peripheral about the status of their request by the CPU. These lines can also act as strobe lines for the requesting devices.

D_0 − D_7

These are bidirectional, data lines which are used to interface the system bus with the internal data bus of DMA controller. In the Slave mode, it carries command words to 8257 and status word from 8257. In the master mode, these lines are used to send higher byte of the generated address to the latch. This address is further latched using ADSTB signal.

IOR

It is an active-low bidirectional tri-state input line, which is used by the CPU to read internal registers of 8257 in the Slave mode. In the master mode, it is used to read data from the peripheral devices during a memory write cycle.

IOW

It is an active low bi-direction tri-state line, which is used to load the contents of the data bus to the 8-bit mode register or upper/lower byte of a 16-bit DMA address register or terminal count register. In the master mode, it is used to load the data to the peripheral devices during DMA memory read cycle.

CLK

It is a clock frequency signal which is required for the internal operation of 8257.

RESET

This signal is used to RESET the DMA controller by disabling all the DMA channels.

A_0 - A_3

These are the four least significant address lines. In the slave mode, they act as an input, which selects one of the registers to be read or written. In the master mode, they are the four least significant memory address output lines generated by 8257.

CS

It is an active-low chip select line. In the Slave mode, it enables the read/write operations to/from 8257. In the master mode, it disables the read/write operations to/from 8257.

A_4 - A_7

These are the higher nibble of the lower byte address generated by DMA in the master mode.

READY

It is an active-high asynchronous input signal, which makes DMA ready by inserting wait states.

HRQ

This signal is used to receive the hold request signal from the output device. In the slave mode, it is connected with a DRQ input line 8257. In Master mode, it is connected with HOLD input of the CPU.

HLDA

It is the hold acknowledgement signal which indicates the DMA controller that the bus has been granted to the requesting peripheral by the CPU when it is set to 1.

MEMR

It is the low memory read signal, which is used to read the data from the addressed memory locations during DMA read cycles.

MEMW

It is the active-low three state signal which is used to write the data to the addressed memory location during DMA write operation.

ADST

This signal is used to convert the higher byte of the memory address generated by the DMA controller into the latches.

AEN

This signal is used to disable the address bus/data bus.

TC

It stands for 'Terminal Count', which indicates the present DMA cycle to the present peripheral devices.

MARK

The mark will be activated after each 128 cycles or integral multiples of it from the beginning. It indicates the current DMA cycle is the 128th cycle since the previous MARK output to the selected peripheral device.

V_{cc}

It is the power signal which is required for the operation of the circuit.

8279

8279 programmable keyboard/display controller is designed by Intel that interfaces a keyboard with the CPU. The keyboard first scans the keyboard and identifies if any key has been pressed. It then sends their relative response of the pressed key to the CPU and vice-a-versa.

How Many Ways the Keyboard is Interfaced with the CPU?

The Keyboard can be interfaced either in the interrupt or the polled mode. In the **Interrupt mode**, the processor is requested service only if any key is pressed, otherwise the CPU will continue with its main task.

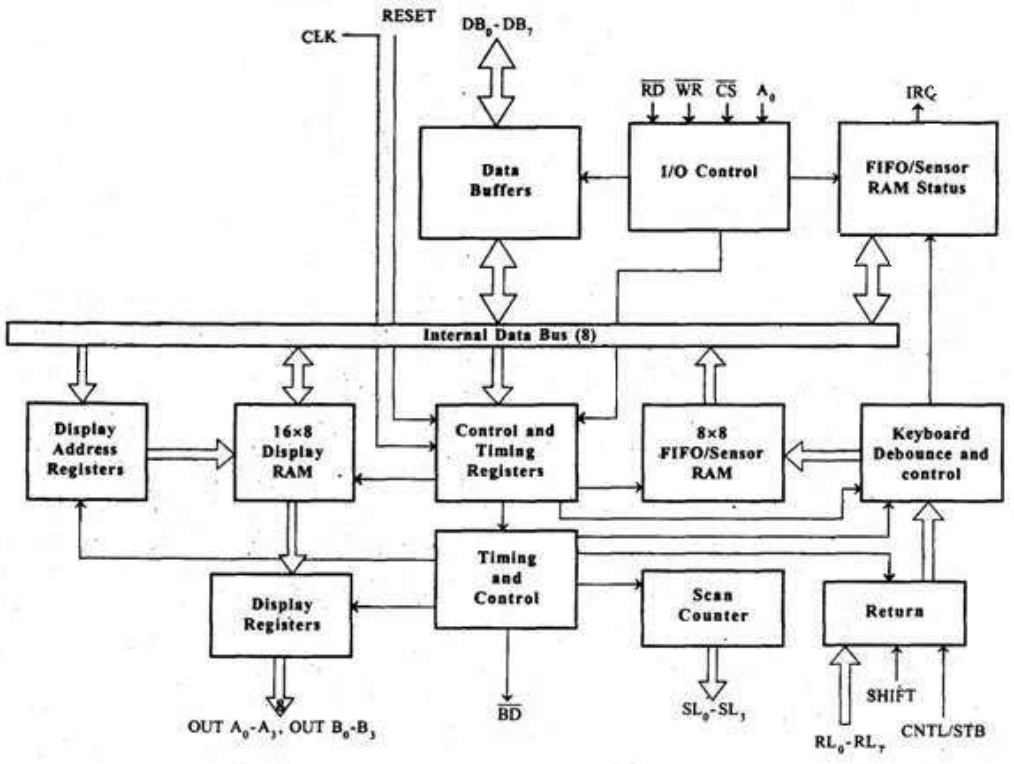
In the **Polled mode**, the CPU periodically reads an internal flag of 8279 to check whether any key is pressed or not with key pressure.

How Does 8279 Keyboard Work?

The keyboard consists of maximum 64 keys, which are interfaced with the CPU by using the key-codes. These key-codes are de-bounced and stored in an 8-byte FIFORAM, which can be accessed by the CPU. If more than 8 characters are entered in the FIFO, then it means more than eight keys are pressed at a time. This is when the overrun status is set.

If a FIFO contains a valid key entry, then the CPU is interrupted in an interrupt mode else the CPU checks the status in polling to read the entry. Once the CPU reads a key entry, then FIFO is updated, and the key entry is pushed out of the FIFO to generate space for new entries.

Architecture and Description



Block diagram of 8279

I/O Control and Data Buffer

This unit controls the flow of data through the microprocessor. It is enabled only when D is low. Its data buffer interfaces the external bus of the system with the internal bus of the microprocessor. The pins A0, RD, and WR are used for command, status or data read/write operations.

Control and Timing Register and Timing Control

This unit contains registers to store the keyboard, display modes, and other operations as programmed by the CPU. The timing and control unit handles the timings for the operation of the circuit.

Scan Counter

It has two modes i.e. **Encoded mode** and Decoded mode. In the encoded mode, the counter provides the binary count that is to be externally decoded to provide the scan lines for the keyboard and display.

In the **decoded scan mode**, the counter internally decodes the least significant 2 bits and provides a decoded 1 out of 4 scan on SL_0-SL_3.

Return Buffers, Keyboard Debounce, and Control

This unit first scans the key closure row-wise, if found then the keyboard debounce unit debounces the key entry. In case, the same key is detected, then the code of that key is directly transferred to the sensor RAM along with SHIFT & CONTROL key status.

FIFO/Sensor RAM and Status Logic

This unit acts as 8-byte first-in-first-out (FIFO) RAM where the key code of every pressed key is entered into the RAM as per their sequence. The status logic generates an interrupt request after each FIFO read operation till the FIFO gets empty.

In the scanned sensor matrix mode, this unit acts as sensor RAM where its each row is loaded with the status of their corresponding row of sensors into the matrix. When the sensor changes its state, the IRQ line changes to high and interrupts the CPU.

Display Address Registers and Display RAM

This unit consists of display address registers which holds the addresses of the word currently read/written by the CPU to/from the display RAM.

8279 − Pin Description

The following figure shows the pin diagram of 8279 −

Data Bus Lines, DB_0 - DB_7

These are 8 bidirectional data bus lines used to transfer the data to/from the CPU.

CLK

The clock input is used to generate internal timings required by the microprocessor.

RESET

As the name suggests this pin is used to reset the microprocessor.

CS Chip Select

When this pin is set to low, it allows read/write operations, else this pin should be set to high.

A_0

This pin indicates the transfer of command/status information. When it is low, it indicates the transfer of data.

RD, WR

This Read/Write pin enables the data buffer to send/receive data over the data bus.

IRQ

This interrupt output line goes high when there is data in the FIFO sensor RAM. The interrupt line goes low with each FIFO RAM read operation. However, if the FIFO RAM further contains any key-code entry to be read by the CPU, this pin again goes high to generate an interrupt to the CPU.

V_{ss}, V_{cc}

These are the ground and power supply lines of the microprocessor.

$SL_0 - SL_3$

These are the scan lines used to scan the keyboard matrix and display the digits. These lines can be programmed as encoded or decoded, using the mode control register.

$RL_0 - RL_7$

These are the Return Lines which are connected to one terminal of keys, while the other terminal of the keys is connected to the decoded scan lines. These lines are set to 0 when any key is pressed.

SHIFT

The Shift input line status is stored along with every key code in FIFO in the scanned keyboard mode. Till it is pulled low with a key closure, it is pulled up internally to keep it high

CNTL/STB - CONTROL/STROBED I/P Mode

In the keyboard mode, this line is used as a control input and stored in FIFO on a key closure. The line is a strobe line that enters the data into FIFO RAM, in the strobed input mode. It has an internal pull up. The line is pulled down with a key closure.

BD

It stands for blank display. It is used to blank the display during digit switching.

$OUTA_0 - OUTA_3$ and $OUTB_0 - OUTB_3$

These are the output ports for two 16x4 or one 16x8 internal display refresh registers. The data from these lines is synchronized with the scan lines to scan the display and the keyboard.

Operational Modes of 8279

There are two modes of operation on 8279 − Input Mode and Output Mode.

Input Mode

This mode deals with the input given by the keyboard and this mode is further classified into 3 modes.

- **Scanned Keyboard Mode** − In this mode, the key matrix can be interfaced using either encoded or decoded scans. In the encoded scan, an 8×8 keyboard or in the decoded scan, a 4×8 keyboard can be interfaced. The code of key pressed with SHIFT and CONTROL status is stored into the FIFO RAM.
- **Scanned Sensor Matrix** − In this mode, a sensor array can be interfaced with the processor using either encoder or decoder scans. In the encoder scan, 8×8 sensor matrix or with decoder scan 4×8 sensor matrix can be interfaced.
- **Strobed Input** − In this mode, when the control line is set to 0, the data on the return lines is stored in the FIFO byte by byte.

Output Mode

This mode deals with display-related operations. This mode is further classified into two output modes.

- **Display Scan** − This mode allows 8/16 character multiplexed displays to be organized as dual 4-bit/single 8-bit display units.
- **Display Entry** − This mode allows the data to be entered for display either from the right side/left side.